Reynold's Transport Theorem

intensive ; extensive

$$\frac{D\Psi}{Dt} = \frac{d}{dt}\left(\iiint \rho \psi \, dV\right) + \iint \rho \psi (\vec{v}\cdot\hat{n})\, dA \qquad ; \quad \psi = \frac{d\Psi}{dm}$$

Material Derivative — Accumulation — Efflux (In - out)

$$\Psi = \{m, P, E\}$$

Mass Balance (Continuity Equation)

$\Psi = m$; $\psi = \frac{d\Psi}{dm} = \frac{dm}{dm} = 1$

$$\frac{Dm}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho \, dV\right) + \iint_{C.S} \rho (\underline{v}\cdot\hat{n})\, dA$$

($\frac{Dm}{Dt} = 0$)

Rate of accumulation of mass w/n CV

Net outward flow of mass across the CS "efflux"

@ S.S = 0

> 0 (Net mass going out)
< 0 (Net mass going in)
= 0 (Mass w/n CV is constant)

Evaluate at all control surfaces of chosen control volume

For incompressible flow: ρ = constant

∴ For SS, incompressible flow through a CV: $\iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA = \frac{dm}{dt}^{0}$



@ SS, incompressible: $\iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA = 0$

$$\rho \iint_{A_1} (\underline{v}\cdot\hat{n})\, dA + \rho \iint_{A_2} (\underline{v}\cdot\hat{n})\, dA = 0$$

$$\rho(-v_1)\iint_{A_1} dA + \rho v_2 \iint_{A_2} dA = 0$$

$$-\rho v_1 A_1 + \rho v_2 A_2 = 0$$

$$\therefore \boxed{\rho v_1 A_1 = \rho v_2 A_2}$$

Irrespective of the flow inside the control volume! (Would work anytime for constant velocity!)

Momentum Balance

$\Psi = m\underline{v}$; $\psi = \frac{d\Psi}{dm} = \frac{d(m\underline{v})}{dm}$

$$\psi = \underline{v}\frac{dm}{dm} + m\frac{d\underline{v}}{dm}^{0}$$

$$\psi = \underline{v}$$

RTT becomes:

$$\frac{D(m\underline{v})}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho \underline{v}\, dV\right) + \iint_{CS} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA$$

From Newton's second Law:

$$\Sigma \underline{F} = \frac{D}{Dt}(m\underline{v})$$

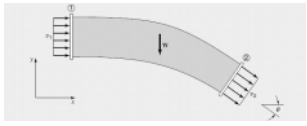
@SS, cartesian coord.

$$\Sigma F_x = \iint \rho v_x (\underline{v}\cdot\hat{n})\, dA$$
$$\Sigma F_y = \iint \rho v_y (\underline{v}\cdot\hat{n})\, dA$$
$$\Sigma F_z = \iint \rho v_z (\underline{v}\cdot\hat{n})\, dA$$

$$\therefore \Sigma \underline{F} = \frac{d}{dt}\left(\iiint_{CV} \rho \underline{v}\, dV\right) + \iint_{C.S.} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA$$

Sum of forces acting on CV i.e. on the fluid — Rate of acc. of momentum w/n CV — net rate of momentum efflux from CV

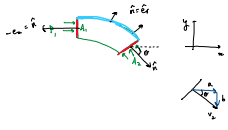
Find the force exerted on a pipe bend resulting from steady state, incompressible flow as shown. W represents gravitational force



① Define CV → All fluid in the pipe

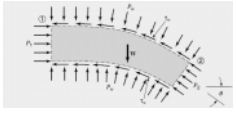
There are external forces acting on this fluid: pressure, τ_w } Total resultant force is defined $\underline{F}$

Force balance:



$\cos\theta = \frac{a}{v_2}$; $\sin\theta = \frac{b}{v_2}$

$\underline{v}_2 = a\hat{i} + b(-\hat{j}) = v_2\cos\theta\,\hat{i} - v_2\sin\theta\,\hat{j}$



Surface	Momentum Balance
①	$\Sigma \underline{F}_1 = \iint_{A_1} \rho \underline{v}(\underline{v}\cdot\hat{n})\, dA_1 = \rho \underline{v} \iint_{A_1} (v_1\hat{i}\cdot(-\hat{i}))\, dA_1 = (-\rho v_1^2 A_1)\hat{i}$
③	$\Sigma \underline{F}_3 = \iint_{A_3} \rho \underline{v}(\underline{v}\cdot\hat{n})\, dA_3$ ($\underline{v}\cdot\hat{n} = 0$)
②	$\Sigma \underline{F}_2 = \iint_{A_2} \rho \underline{v}(\underline{v}\cdot\hat{n})\, dA_2 = \rho \underline{v} \iint (\underline{v}\cdot\hat{n})\, dA_2 = \rho \underline{v}_2 \iint dA_2 = \rho \underline{v}_2 A_2 \begin{pmatrix} v_2\cos\theta \\ -v_2\sin\theta \end{pmatrix} = (\rho v_2^2 A_2 \cos\theta)\hat{i} - (\rho v_2^2 A_2 \sin\theta)\hat{j}$

∴ $\Sigma \underline{F} = \Sigma F_1 + \Sigma F_2$ → Total force acting on the fluid in CV

$$\Sigma \underline{F} = \begin{pmatrix} -\rho v_1^2 A_1 + \rho v_2^2 A_2 \cos\theta \\ -\rho v_2^2 A_2 \sin\theta \end{pmatrix}_{xy}$$

Since the entire system is stationary → net forces are zero

$$\underline{F}_{ext} - \Sigma \underline{F} = 0$$

↓ Taking control volume as entire object

$F_{ext}^{x} = P_1 A_1 - P_2 A_2 \cos\theta + B_x$ ← resultant force on the fluid (don't know the sign)

$F_{ext}^{y} = -W + P_2 A_2 \sin\theta + B_y$

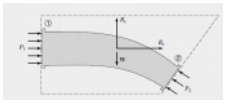
$$\underline{F}_{ext} = \begin{pmatrix} P_1 A_1 - P_2 A_2 \cos\theta + B_x \\ -W + P_2 A_2 \sin\theta + B_y \end{pmatrix}_{xy}$$

$$\begin{pmatrix} P_1 A_1 - P_2 A_2 \cos\theta + B_x - (-\rho v_1^2 A_1 + \rho v_2^2 A_2 \cos\theta) \\ -W + P_2 A_2 \sin\theta + B_y - (-\rho v_2^2 A_2 \sin\theta) \end{pmatrix}_{xy} = 0$$

↓ Solve for B_x, B_y

Now force exerted on pipe by fluid will exert and opposite to force exerted by fluid on pipe

$R_x = -B_x$
$R_y = -B_y$



Reynold's Transport Theorem (RTT)

Intensive / Extensive

$$\frac{D\Psi}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho \psi \, dV\right) + \iint_{CS} \rho \psi (\underline{v}\cdot\hat{n})\, dA \quad ; \quad \psi = \frac{d\Psi}{dm}$$

Material derivative — Accumulation — Efflux (Ins-out)

$\Psi = \{m, \underline{p}, E\}$

Mass Balance (Continuity Equation)

$\Psi = m$; $\psi = \frac{d\Psi}{dm} = \frac{dm}{dm} = 1$ (Replacing $\Psi = m$, $\psi = 1$)

Fluid motion — CV — $\frac{Dm}{Dt} = 0$ — $\frac{\partial m}{\partial t}$ — surface

$$\frac{Dm}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho \, dV\right) + \iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA$$

Rate of accumulation of mass w/in CV — Net outward flow of mass across the CS ('efflux')

@ SS = 0 ✓

- > 0 (Net mass going out)
- < 0 (Net mass going in)
- $= 0$ (Mass w/in CV is constant)

pass from the CV to CS

Evaluate at all control surfaces of chosen control volume

For incompressible flow: ρ = constant

∴ For SS incompressible flow through a CV (No accumulation):

$$\iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA = \frac{\partial m}{\partial t}^{0}$$

$$\frac{\partial m}{\partial t} = \dot{m} = \iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA$$

Fluid is flowing → solve for v_2

$v_2 = f(P, v_1, A)$

@ SS, incompressible

$$\iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA = 0$$

Constant flow (i.e. v_1, v_2)

A_1 A_2

$\underline{v}_1 = v_1 \hat{e}_x$

$\underline{v}_2 = v_2 \hat{e}_x$

$$\iint_{CS} \rho (\underline{v}\cdot\hat{n})\, dA$$

① $\hat{n}_1 = -\hat{e}_x$ ② $\hat{n}_2 = +\hat{e}_x$

$$\Rightarrow \rho \iint_{A_1} (\underline{v}\cdot\hat{n})\, dA + \rho \iint_{A_2} (\underline{v}\cdot\hat{n})\, dA = 0$$

$(\underline{v}\cdot\hat{n}_1) = -v_1$ $(\underline{v}\cdot\hat{n}_2) = +v_2$

$$\Rightarrow \rho \iint_{A_1} (-v_1)\, dA + \rho \iint_{A_2} (+v_2)\, dA = 0$$

v_1 or v_2 $\neq f(A)$ x, y, r, θ

$v_1, v_2 \neq f(x,y)$

$$\Rightarrow \rho(-v_1) \underbrace{\iint_{A_1} dA}_{A_1} + \rho(v_2) \underbrace{\iint_{A_2} dA}_{A_2} = 0$$

$$\int_0^L \int_0^W dx\, dy = \overset{A_1}{LW}$$

$$\boxed{\rho(-v_1)A_1 + \rho v_2 A_2 = 0}$$

$$v_2 = \frac{v_1 A_1}{A_2} \Rightarrow \boxed{v_2 = \frac{v_1 A_1}{A_2}}$$

Derived expression does not concern the flow inside the CV!

Momentum Balance

Momentum $m\underline{v}$ — Fluids → momentum balance

$\Psi = m\underline{v}$ (momentum) ; $\psi = \frac{d\Psi}{dm} = \frac{d(m\underline{v})}{dm} = m\frac{d\underline{v}}{dm} + \underline{v}\frac{dm}{dm} \Rightarrow \boxed{\psi = \underline{v}}$

RTT becomes:

$$\frac{D(m\underline{v})}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho \underline{v}\, dV\right) + \iint_{CS} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA$$

From Newton's 2nd Law:

$$\Sigma \underline{F} = \frac{D(\underline{p})}{Dt} = \frac{D(m\underline{v})}{Dt}$$ → Forces (walls, balls)

$$\Sigma \underline{F} = \frac{d}{dt}\left(\int_V \rho \underline{v}\, dV\right) + \iint_{CS} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA$$

Sum of forces acting on the CV i.e. on the fluid — Rate of acc. of momentum w/in CV — net rate of momentum efflux from CV

@SS, cartesian coord.

$$\begin{cases} \Sigma F_x = \iint \rho v_x (\underline{v}\cdot\hat{n})\, dA \\ \Sigma F_y = \iint \rho v_y (\underline{v}\cdot\hat{n})\, dA \\ \Sigma F_z = \iint \rho v_z (\underline{v}\cdot\hat{n})\, dA \end{cases}$$

or pipe walls — balls, hinges

Find the force exerted on the bend resulting from steady state, incompressible flow as shown. W represents gravitational force.

$\underline{v}_1 = v_1 \hat{e}_x$

① Define CV → All fluid in the pipe

External forces acting on the fluid: P_1, P_2 (opening), $\underline{F}_{CV \to fluid}$, W

Resultant force

RTT → Momentum → forces

These are external forces acting on this fluid: pressure, $\underline{\tau}_w$ — Total resultant force is defined $\underline{F}$

Force balance:

Surface Momentum Balance — SS

$$\Sigma \underline{F} = \iint_{A_1} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA = \rho \underline{v} \iint_{A_1} (\underline{v}\cdot\hat{n})\, dA = \rho \underline{v} \iint_{A_1} v_1 \hat{e}_x \cdot (-\hat{e}_x)\, dA = -\rho \underline{v} v_1 \underbrace{\iint_{A_1} dA}_{A_1} = -\rho \underline{v} v_1 A_1 \Rightarrow \underline{(-\rho v_1^2 A_1)\hat{e}_x} = \Sigma F_1$$

$\hat{e}_x \cdot \hat{e}_x = 1$

$\underline{v} = \underline{v}_1 = v_1 \hat{e}_x$

$\underline{V}_1 = v_1(\hat{e}_x)$ (mag + direction)

$$\Sigma \underline{F}_2 = \iint_{A_2} \rho \underline{v} (\underline{v}\cdot\hat{n})\, dA_2 = \rho \underline{v}_2 \iint_{A_2} (\underline{v}_2\cdot\hat{n})\, dA_2 = \rho \underline{v}_2 (v_2) \underbrace{\iint_{A_2} dA_2}_{A_2} = \rho \underline{v}_2 (v_2) A_2$$

$\underline{v}_2 = v_2 \cos\theta \hat{e}_x + v_2 \sin\theta \hat{e}_y$

$\cos\theta = \frac{a}{v_2}$ $\sin\theta = \frac{b}{v_2}$

Components

$$\Sigma \underline{F}_2 = \rho v_2 A_2 \begin{pmatrix} v_2 \cos\theta \\ -v_2 \sin\theta \end{pmatrix} = \begin{pmatrix} \rho v_2^2 A_2 \cos\theta \\ -\rho v_2^2 A_2 \sin\theta \end{pmatrix}_{xy}$$

$$\therefore \Sigma \underline{F} = \Sigma \underline{F}_1 + \Sigma \underline{F}_2 \rightarrow$$ Total force acting on the fluid in CV

$$\Sigma \underline{F} = \begin{pmatrix} -\rho v_1^2 A_1 + \rho v_2^2 A_2 \cos\theta \\ -\rho v_2^2 A_2 \sin\theta \end{pmatrix}_{xy}$$

Since the entire system is stationary → net forces are zero

$\underline{F}_{ext}$ (force on fluid) + $\underline{F}_{CV \to fluid} - \Sigma \underline{F} = 0$

Taking control volume as entire object

Derived expression does not concern the flow inside the CV !

$\vec{F}_{ext} - \sum \vec{F} = 0$

Taking control volume as entire object



$\vec{F}_{ext} =$

$F_{ext}^x =$ $P_1 A_1 - P_2 A_2 \cos\theta + B_x$ (reactant force (due to P_w, τ))

$F_{ext}^y = -W + P_2 A_2 \sin\theta + B_y$

$F_{ext} =$ $\begin{matrix} P_1 A_1 - P_2 A_2 \cos\theta + B_x \\ -W + P_2 A_2 \sin\theta + B_y \end{matrix}_{xy}$

$\sum F_{ext} = \sum F = 0$ $\Delta P = 0$ → goal is to solve for B_x, B_y

Solve for B_x, B_y

$\begin{matrix} P_1 A_1 - P_2 A_2 \cos\theta + B_x - (-\rho v_1^2 A_1 + \rho v_2^2 A_2 \cos\theta) \\ -W + P_2 A_2 \sin\theta + B_y - (-\rho v_2^2 A_2 \sin\theta) \end{matrix}_{xy} = 0$

solve for B_x, B_y

$P_1 A_1 - P_2 A_2 \cos\theta + \rho v_1^2 A_1 - \rho v_2^2 A_2 \cos\theta = -B_x$

$= -B_y$

Now force exerted on pipe by fluid will exact and opposite to force exerted by fluid on pipe

Force exerted on Pipe R_x, R_y

B_x, B_y → represent force exerted on fluid

equal & opposite

$R_x = -B_x$

$R_y = -B_y$

GH Review Question:

6.47: Steady, incompressible

If pressure no cancel (multiple coordinate dependence)

$$\frac{\partial P}{\partial y} = \alpha$$

$$P = \alpha y + f(x)$$

BC's: @ $y = h$, $P = P_{atm}$

$$\frac{dP}{dx} = \frac{df}{dx}$$

$$P_{atm} = \alpha h + f(x)$$

$$f(x) = \beta = \text{constant}$$

$$\frac{df}{dx} = 0 \Rightarrow \frac{dP}{dx} = 0$$

PROBLEM: 6.47 Flow Problem: Two-layer flow down an incline. Two different fluids with different densities (ρ_1, ρ_2) and viscosities (μ_1, μ_2) are layered on a long plate tilted at an angle β to the horizontal. The thickness of the bottom and more dense fluid is h_1; the thickness of the top and less dense fluid is h_2. The bottom plate is stationary, and a flow is produced by gravity. The flow is steady and both fluids are incompressible. Solve for the velocity profile in a Cartesian coordinate system with flow in the x-direction; y is the direction perpendicular to the plates, with $y = 0$ at the surface of the bottom plate.

x-direction

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial P}{\partial x} + \rho g_x + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right) \quad \text{(E-1)}$$

y-direction

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial P}{\partial y} + \rho g_y + \mu\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right) \quad \text{(E-2)}$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right) \quad \text{(E-3)}$$

Cartesian coordinates:

$$\begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix}_{xyz} = \mu \begin{pmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} & \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \\ \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \\ \frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} & \frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{xyz}$$

Algorithm:

1) Sketch the problem (if no figure provided)
   + Choose coordinate system (or check consistency w/ one provided)
   + Clearly understand where flow is ; where the boundaries are

2) Characterize the flow : $\underline{v} = (\ )$; $\underline{g} = (\ )$
   + Newtonian? Incompressible? Steady state? Where is bulk fluid flowing?
   + Is it open to environment? → Most likely no pressure gradients!
   + Is it inclined? How is gravity affected?

3) Continuity Equation
   + Simplify (and write!) the modified equation (after cancellations)
   + Clearly write why are cancelling out terms

4) Equation of Motion (Microscopic Momentum balance):
   + Incompressible, newtonian (constant viscosity) → Navier stokes
   + Non-newtonian → Cauchy momentum balance

5) Boundary conditions (BC's):
   + No. of BC's correspond to order of differential equation
   + @ solid surfaces → No slip
     @ interfaces → shear stress

6) Solve for velocity profile (and/or pressure field)
   + Integration → keep check of constants
   + solve for constants at very end
   + If pressure is imp (ie. flow in a closed pipe) → then need additional BC's for this

7) Evaluate Q, F given the solved velocity profile

Exam taking strategies

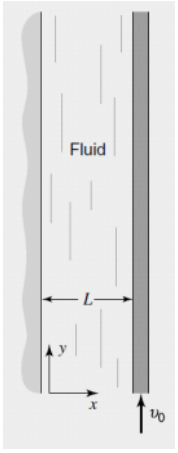
1) Don't panic! → - Read question carefully
   - understand where, what how
   - Check coordinate system !!

2) Maximize points: - Focus on questions w/ higher points (more points ~ more time)
   - Attempt everything & write best guess if short for time (partial credits)

Practice

1) Falling fluid between two plates :

An incompressible fluid confined between two parallel, vertical surfaces. One surface is stationary, whereas the other is moving upward at a constant velocity v0. If we consider the fluid Newtonian and the flow laminar: sketch the velocity streamlines and find the velocity profile in this flow.



+ Cartesian coordinate system (xyz)
+ gravity ✓
+ Newtonian, incomp. ✓
+ Pressure? → closed system so most likely present

$\underline{v} = \begin{pmatrix} 0 \\ -v_y \\ 0 \end{pmatrix}_{xyz}$ $v_y = f(x)$ (Wide/infinite wall i.e. symmetric in z direction)

$\underline{g} = \begin{pmatrix} 0 \\ -g \\ 0 \end{pmatrix}_{xyz}$

→ Mass balance: $0 = \nabla \cdot \underline{v}$ (SS, incomp.)

$\frac{\partial v_x}{\partial x}^{0} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}^{0} = 0$ $\therefore \frac{\partial v_y}{\partial y} = 0$ [Developed flow]

→ EOM: $\rho\left(\frac{\partial \underline{v}}{\partial t}^{SS} + \underline{v} \cdot \nabla \underline{v}\right) = -\nabla p + \mu \nabla^2 \underline{v} + \rho \underline{g}$

x: $0 = -\frac{\partial P}{\partial x}$

y: $0 = -\frac{\partial P}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2}\right) + \rho g_y$; $g_y = -g$

z: $0 = -\frac{\partial P}{\partial z}$

No pressure gradients in x & z direction ; $\frac{\partial P}{\partial x} = \frac{\partial P}{\partial z} = 0$ $P = f(y)$; $\frac{\partial P}{\partial y} = \frac{dP}{dy} = \text{constant}$

$\therefore \frac{1}{\mu}\underbrace{\left(\frac{dP}{dy} - \rho g\right)}_{\text{Not a function of } x!\ = \lambda} = \frac{\partial^2 v_y}{\partial x^2}$

$\frac{\partial v_y}{\partial x} = \lambda x + C_1$

Not a function of x!
$= \lambda$

$$\frac{\partial v_y}{\partial x} = \frac{\lambda}{\mu} x + C_1$$

$$v_y = \frac{\lambda}{\mu}\frac{x^2}{2} + C_1 x + C_2$$

Boundary conditions:

@ $x = 0$, $v_y = 0$

@ $x = L$, $v_y = v_0$

$C_2 = 0$

$$\left(v_0 - \frac{\lambda}{\mu}\frac{L^2}{2}\right)\frac{1}{L} = C_1$$

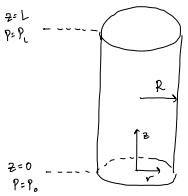
$$C_1 = \frac{v_0}{L} - \frac{\lambda L}{2\mu}$$

$$\boxed{v_y = \frac{\lambda}{\mu}\frac{x^2}{2} + \left(\frac{v_0}{L} - \frac{\lambda L}{2\mu}\right)x}$$

Sketch it out !

2) 6.44 & 6.52

Upward pressure-driven flow in a tube ; Non-newtonian fluid (w/ power law) ; Incompressible ; SS



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{rθz}$

$\underline{g} = \begin{pmatrix} 0 \\ 0 \\ -g \end{pmatrix}_{rθz}$

Flow profile?

Pressure profile?

Cylindrical coordinates

→ Mass balance : $\nabla \cdot \underline{v} = 0$ (incomp., ss.)

$$\cancel{\frac{\partial \rho}{\partial t}}^{\,0} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r \cancel{v_r}^{\,0}) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho \cancel{v_\theta}^{\,0}) + \frac{\partial}{\partial z}(\rho v_z) = 0$$

$$\frac{\partial v_z}{\partial z} = 0$$

→ Non-newtonian flow → Cannot use Navier stokes!
(only incompressible flows)

Table B.6 : Cylindrical coordinates of Cauchy momentum balance

$\tau = \eta(\dot\gamma)\,\dot{\underline{\gamma}}$

η for non-newt. ↓ μ

$\dot{\underline{\gamma}} = \left(\nabla \underline{v} + (\nabla \underline{v})^T\right) = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix}$

$\eta(\dot\gamma) = m\dot\gamma^{n-1}$

Equation 8.8-2 :

Newtonian constitutive equation :

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix} = \eta(\dot\gamma) \underbrace{\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}}_{\dot{\underline{\gamma}}}$$

$$\tau_{rz} = \frac{\partial v_z}{\partial r}$$

$$\dot\gamma = \|\dot{\underline{\gamma}}\| = \left\| \begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} \right\|$$

$$= \sqrt{\frac{1}{2}\left(\text{sum of squares of each term in } \dot{\underline{\gamma}}\right)}$$

$$= \sqrt{\frac{1}{2}\left[\left(\frac{\partial v_z}{\partial r}\right)^2 + \left(\frac{\partial v_z}{\partial r}\right)^2\right]} = \left|\frac{\partial v_z}{\partial r}\right|$$

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

as $r\uparrow$, $v_z\downarrow$ (closer to the wall)

$$\frac{\partial v_z}{\partial r} < 0 \quad \rightarrow \quad -\frac{\partial v_z}{\partial r} > 0$$

$$\therefore \quad \dot{\gamma} = -\frac{\partial v_z}{\partial r}$$

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix} = \eta(\dot{\gamma}) \underbrace{\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{\partial v_\theta}{\partial z} + \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{sym}}_{\dot{\gamma}}$$

$$= m\dot{\gamma}^{n-1} \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$= m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1} \begin{pmatrix} 0 & 0 & \partial v_z/\partial r \\ & 0 & 0 \\ & & 0 \end{pmatrix}$$

$$\therefore \quad \tau_{rz} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) \neq f(\theta, z)$$

Only τ_{rz} exists (due to flow direction) which is a function of r (and not z!)

r-direction: $0 = -\frac{\partial P}{\partial r}$

θ-direction: $0 = -\frac{1}{r}\frac{\partial P}{\partial \theta}$

z-direction: $0 = -\frac{\partial P}{\partial z} + \left(\frac{1}{r}\frac{\partial}{\partial r}(r\cdot\tau_{rz})\right) + \underbrace{\rho g_z}_{-fg}$

P has no θ & r dependence

$-\frac{\partial P}{\partial z} = -\frac{dP}{dz}$

$$\therefore \quad \underbrace{\frac{dP}{dz} + \rho g}_{\neq f(r)} = \underbrace{\frac{1}{r}\frac{d}{dr}(r\tau_{rz})}_{\neq f(z)} = \lambda$$

↳ Solve them separately!

↳ $\frac{dP}{dz} + \rho g = \lambda$

$\frac{dP}{dz} = \lambda - \rho g$

$P = (\lambda - \rho g)z + c_1$

@ $z = 0$, $P = P_0$ } $c_1 = P_0$
@ $z = L$, $P = P_L$ }

$(P_L - P_0) = (\lambda - \rho g)L$

$\lambda = \frac{P_L - P_0}{L} + \rho g$

↳ $\frac{d}{dr}(r\tau_{rz}) = \lambda r$

$\tau_{rz} = \frac{\lambda r}{2} + \frac{c_1}{r}$

@ $r = 0$; $\frac{\partial v_z}{\partial r} = 0$; $\tau_{rz} = 0$ } $c_1 = 0$

$\tau_{rz} = \frac{\lambda r}{2}$

↳ $\tau_{rz} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) \times \frac{(-1)}{(-1)}$

$\tau_{rz} = -m\left(-\frac{\partial v_z}{\partial r}\right)^n$

$\tau_{rz} = \frac{\lambda r}{2} = -m\left(-\frac{\partial v_z}{\partial r}\right)^n$

$\downarrow$

$-\frac{dv_z}{dr} = \left(-\frac{\lambda}{2m}\right)^{\frac{1}{n}} r^{1/n}$

$v_z = -\left(-\frac{\lambda}{2m}\right)^{1/n} \frac{r^{1/n+1}}{\left(\frac{1}{n}+1\right)} + c_1$

@ $r = R$, $v_z = 0$ } $c_1 = \left(-\frac{\lambda}{2m}\right)^{1/n} \frac{R^{(\frac{1}{n}+1)}}{\left(\frac{1}{n}+1\right)}$

$$\therefore \quad \boxed{v_z = \left(-\frac{\lambda}{2m}\right)^{1/n} \frac{1}{\left(\frac{1}{n}+1\right)}\left(R^{\frac{1}{n}+1} - r^{\frac{1}{n}+1}\right)}$$

Algorithm:

1) Sketch the problem (if no figure provided)
   + Choose coordinate system (or check consistency w/ one provided)
   + Clearly understand where flow is; where the boundaries are
2) Characterize the flow: $\underline{v} = (\ )$; $\underline{g} = (\ )$
   + Newtonian? Incompressible? Steady state? Where is both fluid flowing?
   + Is it open to environment? → Most likely no pressure gradient!
   + Is it inclined? How is gravity affected?
3) Continuity Equation (Mass balance)
   + Simply (and write!) the modified equation (after cancellations)
   + Clearly write why are cancelling out terms
4) Equation of Motion (Microscopic Momentum balance):
   + Incompressible, newtonian (constant viscosity) → Navier stokes
   + Non-newtonian → Cauchy momentum balance
5) Boundary conditions (BC's):
   + No. of BC's correspond to order of differential equation
   + @ solid surfaces → No slip } @ rest $v=0$, @ moving $v = v_{surface}$
   @ interfaces → shear stress ; l-l ; l-air
6) Solve for velocity profile (and/or pressure field)
   + Integration → keep check of constants
   + solve for constants at very end @ BC's
   + If pressure is imp (i.e. flow in a closed pipe) → then need additional BC's for this
7) Evaluate Q, F given the solved velocity profile } Exam 1

flow forces

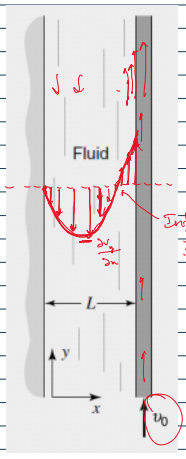
Exam taking strategies

1) Don't panic! → - Read question carefully
   - understand where what has
   - Check coordinate system !! (right hand)
2) Maximize points: - Focus on questions w/ higher points (more points ~ more time)
   - Attempt everything & write "best guess" if short for time (partial credits)

Practice

1) Falling fluid between two plates:

An incompressible fluid confined between two parallel, vertical surfaces. One surface is stationary, whereas the other is moving upward at a constant velocity v0. If we consider the fluid Newtonian and the flow laminar: sketch the velocity streamlines and find the velocity profile in this flow.



+ Cartesian coordinate system (xyz)
+ gravity
+ Newtonian, incomp.
+ Pressure? → closed system so most likely present

$\underline{v} = \begin{pmatrix} 0 \\ v_y \\ 0 \end{pmatrix}_{xyz}$; $v_y = f(x)$ (Wide/infinite wall i.e. symmetric in z direction)

$\underline{g} = \begin{pmatrix} 0 \\ -g \\ 0 \end{pmatrix}_{xyz}$

→ Mass balance: $0 = \nabla \cdot \underline{v}$ (SS, incomp.)

$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$; $\therefore \frac{\partial v_y}{\partial y} = 0$ [Developed flow]

→ EOM: Navier

$$\rho\left(\frac{\partial \underline{v}}{\partial t} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \mu\nabla^2\underline{v} + \rho\underline{g}$$

→ x: $0 = -\frac{\partial P}{\partial x}$

→ y: $0 = -\frac{\partial P}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2}\right) + \rho g_y$; $g_y = -g$

→ z: $0 = -\frac{\partial P}{\partial z}$

No pressure gradients in x & z direction; $\frac{\partial P}{\partial x} = \frac{\partial P}{\partial z} = 0$; $P = f(y)$; $\frac{\partial P}{\partial y} = \frac{dP}{dy} = \text{constant}$ (no x-dependence)

$$\therefore \frac{1}{\mu}\left(\frac{dP}{dy} - \rho g\right) = \frac{\partial^2 v_y}{\partial x^2}$$ (Not a function of x) → $\frac{\partial^2 v_y}{\partial x^2} = \text{constant}$

$\frac{\partial v_y}{\partial x} = \lambda x + C_1$

Not a function of z!

$$\frac{\partial v_z}{\partial r} = \frac{\lambda}{\mu} r + c_1$$

$$v_z = \frac{\lambda}{\mu}\frac{r^2}{2} + c_1 r + c_2$$

BC's

Boundary conditions:

@ $r = 0$, $v_z = 0$ ✓ → $c_2 = 0$

@ $r = L$, $v_z = v_0$ ✓ → $\left(v_0 - \frac{\lambda}{\mu}\frac{L^2}{2}\right)\frac{1}{L} = c_1$

$$c_1 = \frac{v_0}{L} - \frac{\lambda L}{2\mu}$$

$$v_z = \frac{\lambda}{\mu}\frac{r^2}{2} + \left(\frac{v_0}{L} - \frac{\lambda L}{2\mu}\right) r$$

$v \sim r$

r^2 parabolic

Sketch it out!

Wall on r, down $r=0$, $r=L$

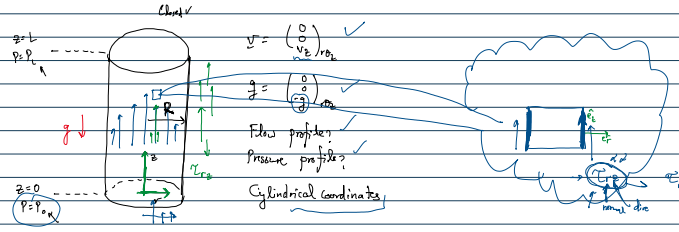
v_z ✓

2) 6.44 & 6.52

Upward pressure-driven flow in a tube; Non-newtonian fluid (w/ power law); Incompressible; SS

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters (m) and (n).



→ Mass balance: $\nabla \cdot \underline{v} = 0$ (incomp., ss.)

$$\frac{\partial \rho}{\partial t}^{0\,(ss)} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r)^{0} + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta)^{0} + \frac{\partial}{\partial z}(\rho v_z) = 0$$

$\Rightarrow \rho \frac{\partial v_z}{\partial z} = 0$

(developed flow) $\frac{\partial v_z}{\partial z} = 0$

→ Non-newtonian flow → Cannot use Navier stokes!
(only incompressible flows)

Table B.6: Cylindrical coordinates of Cauchy momentum balance

$\tau = \mu(\nabla \underline{v} + (\nabla \underline{v})^T)$; $\underline{\dot{\gamma}}$

$\underline{\tau} = \eta(\dot{\gamma})\,\underline{\dot{\gamma}} = \eta(\dot{\gamma}) \left[\nabla \underline{v} + (\nabla \underline{v})^T\right]$

shear strain deformation tensors

$\eta(\dot{\gamma})$ for non-newt.; μ for newt.

$\eta(\dot{\gamma}) = m\dot{\gamma}^{n-1}$; $\eta(\dot{\gamma})$

$\underline{\tau} = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix}$

$m(\dot{\gamma})^{n-1}$

$v_r = v_\theta = 0$, v_z

$r\theta_z$

Equation 8.8-2:

Newtonian constitutive equation:

$\underline{\tau} = \eta(\dot{\gamma})\,\underline{\dot{\gamma}}$

$\eta \cdot \mu$

$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{pmatrix} = \eta(\dot{\gamma}) \begin{pmatrix} 2\frac{\partial v_r}{\partial r}^{0} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)^{0} + \frac{1}{r}\frac{\partial v_r}{\partial \theta}^{0} & \frac{\partial v_r}{\partial z}^{0} + \frac{\partial v_z}{\partial r} \\ \cdot & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}^{0} + \frac{v_r}{r}^{0}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta}^{0} + \frac{\partial v_\theta}{\partial z}^{0} \\ \cdot & \cdot & 2\frac{\partial v_z}{\partial z}^{0} \end{pmatrix}$

$r\theta_z$

↳ τ → shear stress

$\tau_{rz} = \frac{\partial v_z}{\partial r}$

$\underline{\dot{\gamma}} = \begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix}$

$\dot{\gamma} = \|\underline{\dot{\gamma}}\|$ = magnitude

$= \sqrt{\frac{1}{2}\left(\text{sum of squares of each term in } \underline{\dot{\gamma}}\right)}$

$= \sqrt{\frac{1}{2}\left[\left(\frac{\partial v_z}{\partial r}\right)^2 + \left(\frac{\partial v_z}{\partial r}\right)^2\right]} = \left|\frac{\partial v_z}{\partial r}\right|$

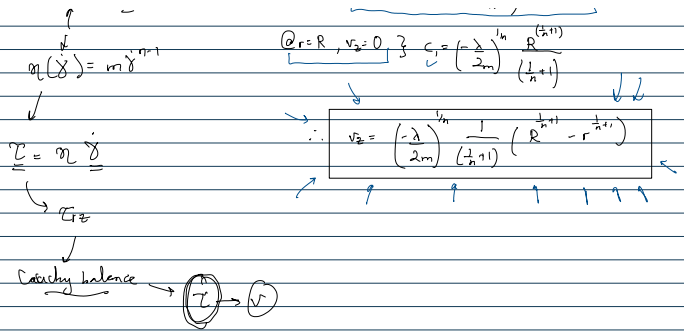
+ or (−)

1) Non-newton:

$\underline{\tau} = \eta\,\underline{\dot{\gamma}}$

2) $\eta = f(\dot{\gamma}) = m(\dot{\gamma}^{n-1})$

Reviews Page 12


$\eta(\dot{\gamma}) = m\dot{\gamma}^{n-1}$
$\underline{\tau} = \eta\,\underline{\dot{\gamma}}$
τ_{rz}
Cauchy balance
τ
v
@ $r = R$, $v_z = 0$ ⇒ $C_1 = \left(\frac{\lambda}{2m}\right)^{1/n} \frac{R^{\left(\frac{1}{n}+1\right)}}{\left(\frac{1}{n}+1\right)}$
$\therefore v_z = \left(\frac{\lambda}{2m}\right)^{1/n} \frac{1}{\left(\frac{1}{n}+1\right)} \left(R^{\frac{1}{n}+1} - r^{\frac{1}{n}+1}\right)$

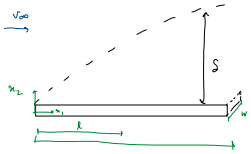
Boundary Layer

Laminar [$Re_{x_1} < 2\times10^5$]

+ No slip condition (gives rise to flow dependent on two directions)

$$\delta = \left(\frac{25\mu}{\rho v_\infty}\right)\sqrt{l} = \frac{5x}{\sqrt{Re}}$$

(l: distance from leading edge of the plate [Not a var!])



Smooth flat plate BL

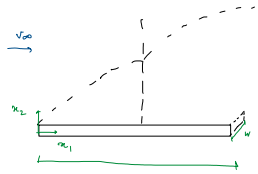
Turbulent [$Re_{x_1} > 1\times10^6$]

+ Downstream flow (from the leading edge) the velocity becomes unstable } i.e. only starts after a initial laminar regime BL

+ Thicker layers (that have different x_1 dependence) } Higher drag on flat plate

$$\delta = \frac{0.16l}{(Re)_x^{1/7}} \quad ; \quad C_D = \frac{0.031}{(Re_L)^{1/7}} \quad ; \quad C_f = \frac{0.027}{(Re_x)^{1/7}} = \frac{2\tau_w}{\rho U_0^2} \quad ; \quad U_0 = v_\infty$$

$$\tau_w = \frac{0.0135\,\rho v_\infty^2}{(Re_x)^{1/7}}$$



Smooth flat plate BL

Laminar (upto a certain x_1) | turbulent

Fluid force on plate:

$$F = \iint_S [\hat{n}\cdot\pi]|_{surf}\,dS \longrightarrow \text{pg 693} \longrightarrow \tilde{\tau}_w = 0.332\,\mu v_\infty \left(\frac{\rho v_\infty}{\mu l}\right)^{1/2}$$

$$F_{drag} = \int_0^W \int_0^L \tilde{\tau}_w\,dx_1\,dx_3 \quad ; \quad Re = \frac{\rho v_\infty l}{\mu}$$

★

$$C_D = \frac{F_{drag}}{\frac{1}{2}\rho v_\infty^2 WL} = \frac{1.328}{(Re)_L^{1/2}}$$

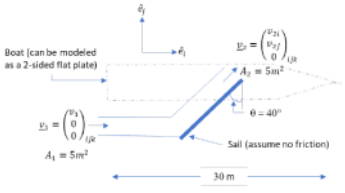
General relationship: $C_D = \dfrac{F_{drag}}{(KE)(Area)} = \dfrac{\iint \tau_w \cdot dA}{(KE)(Area)} = \dfrac{\iint C_F \cdot \left(\frac{\rho v_\infty}{2}\right) \cdot dA}{(KE)(Area)}$

(C_F: dependent on lam/turb } dependent on x var)

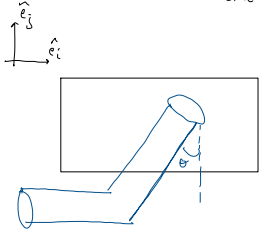
} Use C_f Unless evaluating for entire plat BL!)

1. Come Sail Away

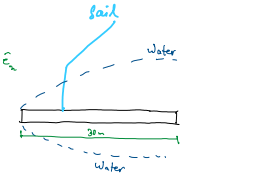
You are captain on a sailboat. Your boat is moving completely due to the force of the wind in the sail. The wind enters the sail region through an area that is roughly 5 m², with a velocity $\underline{v}_1 = \begin{pmatrix} v_1 \\ 0 \\ 0 \end{pmatrix}_{ijk}$. The wind leaves the sail region through an area that is also 5 m², with a velocity $\underline{v}_2 = \begin{pmatrix} v_{2i} \\ v_{2j} \\ 0 \end{pmatrix}_{ijk}$. Your boat is 30m long, and it can be modeled as a 2-sided flat plate. You can consider that the boat extends 2 m below the surface of the water. On this day, your boat is moving at a constant velocity of 0.5 m/s. The water has a density of 1000 kg/m³ and a viscosity of 9x10⁻³ poise. The viscosity of air is 1.7x10⁻⁵ poise, and the density of air is 1.2 kg/m³.



[Boat]
Two-sided flat plate;

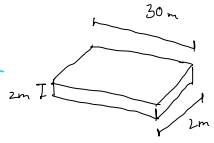


Side view of the boat: →



"Kinda like a hydrofoil of a sailboat" I.e. submerged.





1 poise = 0.1 $\frac{kg}{m\cdot s}$

Water BL: $Re_L = \dfrac{\rho v_\infty L}{\mu}$

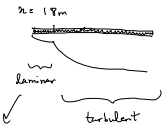
$= \dfrac{(1000)(0.5)(30)}{1.176\times10^{-7}}$ (turbulent)

Water BL: $Re_L = \frac{\rho v_\infty L}{\mu}$

$= \frac{(1000)(0.5)(30)}{(0.9)\times 10^{-3}} = 1.67\times 10^7$ (turbulent)

Finding transition to turbulent: $1\times 10^6 = \frac{(1000)(0.5)(L)}{(0.9\times 10^{-3})}$

$L = 1.8$ m



$\tau_w^l = \frac{0.332\,\rho v_\infty^2}{(Re_x)^{1/2}}$ $\quad \tau_w^t = \frac{0.0135\,\rho v_\infty^2}{(Re_x)^{1/7}}$

$F_D = \iint \tau_w^l \cdot dx\,dy + \iint \tau_w^t\, dx\,dy$

$= \int_0^2\int_0^{1.8} \frac{(0.332)\,\rho v_\infty^2}{\left(\frac{\rho v_\infty x}{\mu}\right)^{1/2}} dx\,dy + \int_0^2\int_{1.8}^{30} \frac{0.0135\,\rho v_\infty^2}{\left(\frac{\rho v_\infty x}{\mu}\right)^{1/7}} \cdot dx\,dy$ $\quad \frac{6}{7}-\frac{1}{2} = \frac{12}{14}$, $\quad -\frac{1}{7}+\frac{1}{14}$

$= 2\left(\int_0^{1.8} 0.332\,\rho^{1/2} v_\infty^{2/2} \mu^{1/2} x^{-1/2} \cdot dx\right) + 2\left(\int_{1.8}^{30} 0.0135\,\rho^{6/7} v_\infty^{13/7} \mu^{1/7} x^{-1/7}\, dx\right)$

$= 2(0.332)\sqrt{\rho\mu}\,(v_\infty)^{2/2}\,\frac{x^{1/2}}{(1/2)}\Big|_0^{1.8} + 2(0.0135)\,\rho^{6/7} v_\infty^{13/7} \mu^{1/7}\,\frac{x^{6/7}}{(6/7)}\Big|_{1.8}^{30}$

$F_D = \left[2(0.332)\sqrt{(1000)(0.9\times 10^{-3})}\cdot(0.5)^{2/2}\,\frac{(1.8)^{1/2}}{(1/2)}\right] + \left[2(0.0135)(1000)^{6/7}(0.5)^{13/7}(0.9\times 10^{-3})^{1/7}\,\frac{30^{6/7}-(1.8)^{6/7}}{(6/7)}\right]$

Force of drag from just 1 BL

$= 0.5976 + 19.99$

$F_D = 20.6$ N

Total F_D (from both BL) $\Rightarrow 2F_D = F_D^{total} \Rightarrow \boxed{F_D^{total} = 41.2\ N}$

Evaluate the velocities using momentum balance:

$\beta = 1$ (turbulent); $\frac{dp}{dt} = 0$ + $\sum \rho \cos\theta_i A_i \langle v_i\rangle^2 \hat{v}_i = \sum f = \underbrace{\underline{f}_g + \underline{f}_{in}}_{\text{negligible}} + \underline{f}_{out} + \underline{f}_w$; $f_w = -F_D^{total}$

$\rho\cos\theta_1 A_1 \langle v_1\rangle^2 \hat{v}_1 + \rho\cos\theta_2 A_2 \langle v_2\rangle^2 \hat{v}_2 = -41.2$ N

$\rho A\left[(-1)\langle v_1\rangle^2 \hat{e}_n + (1)\langle v_2\rangle^2 \begin{pmatrix}\cos 50\\ \sin 50\end{pmatrix}\right] = -41.2$ N

Only care about i^{th} direction:

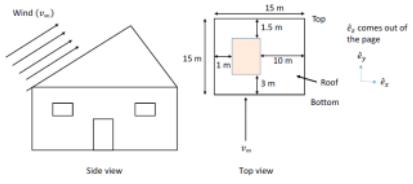
$\Rightarrow \rho A\left[-\langle v_1\rangle^2 + \langle v_2\rangle^2 \cos 50\right] = -41.2$; $v_1 = v_2 = (v_1 - 0.5)$ ← Finding the relative velocity (since boat is moving as well)

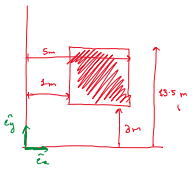
$\Rightarrow \rho A\left[-(v_1-0.5)^2 + (v_1-0.5)^2\cos 50\right] = -41.2$

$(v_1-0.5)^2 \rho A\,(-1+\cos(50)) = -41.2$

$\boxed{v_1 = 4.88\ m/s}$

Raise the Roof





Given an Fdrag $F_D = C_f \cdot F_g$; $F_g = m \cdot g$

$16 = 3+2+1.5$
$x = 5+4.5$
$= 10.5$

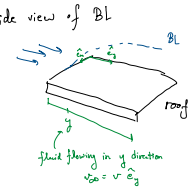
We don't know the v_∞ → have to assume a regime since we cannot find Re

Assuming turbulent on the board (i.e. laminar regime has been crossed before the board ↓ But we don't care)

$$F_D = \iint \tau_w \cdot dx\, dy \quad ; \quad \tau_w = \frac{\rho v_\infty^2}{2} C_f = \frac{0.0135\, \rho v_\infty^2}{(Re_x)^{1/7}} \quad ; \quad Re_y = \frac{\rho v_\infty y}{\mu}$$

Boundary layer is developing in the y direction → that is the characteristic dimension now (given the coordinate system we have)

Side view of BL



↳ Given exam solution are also technically correct but they changed the coordinate axis given arbitrarily

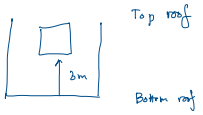
$$F_D = \int_3^{13.5} \int_1^5 0.0135\, \rho^{6/7} v_\infty^{13/7} \mu^{1/7} (y)^{-1/7}\, dx\, dy$$

$$= (5)\; 0.0135\, \rho^{6/7} v_\infty^{13/7} \mu^{1/7} \left. \frac{(y)^{(6/7)}}{(6/7)} \right|_3^{13.5}$$

$$C_f \cdot (F_g) = v_\infty^{13/7} \left[(5)(0.0135) \left[\frac{1}{3} \right]^{6/7} (1.7\times10^{-5})^{1/7} \left[\frac{13.5^{6/7} - 3^{6/7}}{6/7} \right] \right]$$

$$(1.2)(70) = v_\infty^{13/7} (0.1385)$$

$$v_\infty = \left(\frac{(1.2)(70)}{0.1385} \right)^{7/13}$$



$$\boxed{v_\infty = 31.5 \approx 32 \text{ m/s}}$$

↳ Assumption check: @ start of board ↓

$$Re = \frac{(1.3)(31.5)(3)}{(1.7\times10^{-5})} > 1\times10^6$$

✓ Board is in turbulent

$\delta^{turb} = \frac{0.16 \cdot x}{(Re_x)^{1/7}}$ → x @ which we want to measure the thickness (δ changes w/ x!)

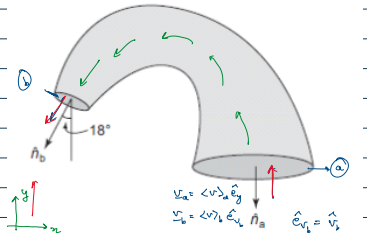
∴ @ x = 14.5 (the birds nest)

$$\delta^{turb} = \frac{(0.16)(14.5)}{\left(\frac{\rho v_\infty (14.5)}{\mu}\right)^{1/7}} = \frac{(0.16)(14.5)}{\left[\frac{(1.3)(31.5)(14.5)}{(1.7\times10^{-5})}\right]^{1/7}} = 0.194 \text{ m}$$

Nest is 0.25 m tall ; $\delta_{turb} = 0.19$ m

It is not in the BL & experiences v_∞ (will likely 'blow away)!

Question 1



A liquid with a given viscosity (mew) enters and leaves the above bend as shown. Solve the following questions

$\langle \underline{v}_a \rangle = \frac{\iint \underline{v}\, dA}{\int dA}$ → Mass Flow Rate

$\underline{v}_a = f(r)\, \hat{e}_y$

RTT → $\iint \underline{v} \cdot \rho (\underline{v} \cdot \hat{n})\, dA$

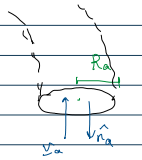
(a) Using Reynolds transport theorem, solve for the overall force on the fluid.

RTT: $\frac{d\underline{P}}{dt} + \iint_{CS} \rho (\underline{v} \cdot \hat{n}) \underline{v}\, dS = \sum_{on\ CV} \underline{F}$

Taking fluid inside the bend as CV: surface (A) & (B) are the flow surfaces

(A) → Inlet; (B) → outlet

Inlet:



$\underline{v}_a = \begin{pmatrix} 0 \\ \langle v \rangle_a \\ 0 \end{pmatrix}_{xyz}$; $\hat{n}_a = -\hat{e}_y = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}_{xyz}$

$\underline{v} = \langle \underline{v} \rangle_a \hat{e}_y = \left(\frac{r^2}{2} + 4r\right) \hat{e}_r = \begin{pmatrix} \frac{r^2}{2} + 4r \\ 0 \\ 0 \end{pmatrix}$ $r\theta z$ → xyz

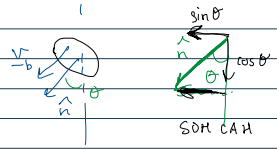
$\iint_{inlet} \rho \underline{v}_a (\hat{n} \cdot \underline{v}_a)\, dS = \iint \rho (\langle v \rangle_a \hat{e}_y) \left[\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}_{xyz} \cdot \begin{pmatrix} 0 \\ \langle v \rangle_a \\ 0 \end{pmatrix}_{xyz} \right] \cdot dS$

$= \int_0^{2\pi} \int_0^{R_a} \rho \langle v \rangle_a \hat{e}_y \left[-\langle v \rangle_a \right] r\, dr\, d\theta$

$= -\rho \langle v \rangle_a^2 \hat{e}_y \int_0^{2\pi} \int_0^{R_a} r\, dr\, d\theta$

$= \left(-\rho \langle v \rangle_a^2 \pi R_a^2 \right) \hat{e}_y = \begin{pmatrix} 0 \\ -\rho \langle v \rangle_a^2 \pi R_a^2 \\ 0 \end{pmatrix}_{xyz}$ (a)

Outlet:



$\hat{n}_b = \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz}$; $\hat{e}_{v_b} = \begin{pmatrix} \sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz}$ → $\underline{v}_b = \langle v \rangle_b \hat{e}_{v_b} = \langle v \rangle_b \begin{pmatrix} \sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$

$\iint_{C.S} \rho (\underline{v} \cdot \hat{n}) \underline{v}\, dS = \int_0^{2\pi} \int_0^{R_b} \rho \left(\langle v \rangle_b \hat{e}_{v_b} \right) \left[\begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz} \cdot \begin{pmatrix} -\langle v_b \rangle \sin\theta \\ -\langle v_b \rangle \cos\theta \\ 0 \end{pmatrix} \right] r\, dr\, d\theta$

(a) $= \rho \langle v \rangle_b^2 \pi R_b^2 \hat{e}_{v_b}$ θ ✓

$\begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz}$ (b)

RTT

$\sum_{on\ CV} \underline{F} = \iint_{CS} \rho (\underline{v} \cdot n) \underline{v}\, dS$ = outlet + inlet

$\sum \underline{F} = -\rho \pi R_a^2 \langle v \rangle_a^2 \begin{pmatrix} 0 \\ +1 \\ 0 \end{pmatrix} + \rho \pi R_b^2 \langle v \rangle_b^2 \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$ (b)

Exam 3 → Macro Momentum

(b) Using macroscopic momentum balance, solve for the overall force on the fluid

(b) Using macroscopic momentum balance, solve for the overall force on the fluid

$$\sum F_{on\,CV} = \frac{dP}{dt} + \sum_i \left[\frac{\rho A_i \cos\theta_i \langle v \rangle_i^2}{\beta} \right] \hat{v}_i$$

$$= \left. \frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right|_{inlet} + \left. \frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right|_{outlet}$$

$$= \frac{\rho \pi R_a^2 (\cos 180^\circ) \langle v \rangle_a^2}{\beta} \hat{v}_a + \frac{\rho \pi R_b^2 \cos(0) \langle v \rangle_b^2}{\beta} \hat{v}$$

$\beta = 1$ (turbulent)

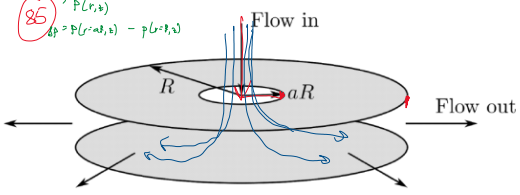
$$\sum F = -\rho \pi R_a^2 \langle v \rangle_a^2 \begin{pmatrix} 0 \\ +1 \\ 0 \end{pmatrix} + \rho \pi R_b^2 \langle v \rangle_b^2 \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$$

(c) Are your force expressions in (a) and (b) different from each other? Why or why not?

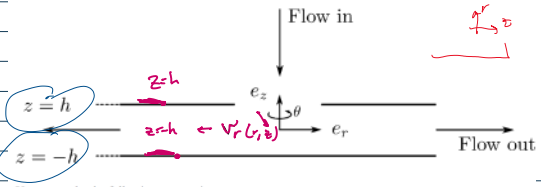
Same, because

Question 2

Consider two concentric disks which are fixed to stay in place. There is an inflow of an incompressible fluid through a hole in the top plate. The flow is driven by a pressure drop, $\Delta P = p(r = aR) - p(r = R)$, between the inlet at $r = aR$ ($a \ll 1$) and the outlet at $r = R$. Below, you will find the 3D view of the apparatus.



The following figure shows the front view of the apparatus above. Consider the cylindrical coordinate system to be positioned in the center.



You can make the following assumptions:

1. The outflow is homogeneous along the circumference.
2. The flow is fully developed and at steady-state.
3. The fluid is incompressible.
4. The outlet is at atmospheric pressure, p_{atm}.
5. The effects due to flow around the corner at the inlet can be neglected ($a \ll 1$). The pressure at that point is unperturbed by the flow around the corner.

(a) (10 points).
Write down the kinematics for this problem by stating physically reasonable assumptions.
(i) Which velocity components are non-zero?
(ii) Which coordinates (i.e. r, θ, z) do the non-zero velocity component(s) depend on?

(i) $\underline{v} = \begin{pmatrix} v_r \\ v_\theta \\ v_z \end{pmatrix}_{r\theta z}$; $v_\theta = v_z = 0$

$\underline{v} = \begin{pmatrix} v_r \\ 0 \\ 0 \end{pmatrix}_{r\theta z}$

(symmetry)

$f(r, \theta, z)$

v_z

$\underline{v} = \begin{pmatrix} v_r \\ 0 \\ 0 \end{pmatrix}_{r\theta z}$

(symmetry)

(ii) $v_r = f(r, \theta, z)$

(b) (20 points) micro macro velo

Use the local form of the balance of mass to determine the radial component of the velocity profile in terms of an integration constant. Write your final answer in the box provided below.

$\nabla \cdot \underline{v} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$

Mass balance : @ SS, incom $\nabla \cdot \underline{v} = 0$ (polar coordinates)

$\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) = 0$; $v_\theta = v_z = 0$

$\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) = 0$; ρ = constant

$\frac{\partial}{\partial r}(r v_r) = 0$ Integrate once

$r v_r = C_1$

$v_r = \frac{C_1}{r}$; $C_1 = f(z)$

$v_r = f(r, z)$

$v_r = \frac{C_1}{r}$ (i)

$z \neq f(x)$

$y = 4x + z$

$\frac{\partial y}{\partial z} = 4$

der

$\frac{\partial (r v_r)}{\partial r} = \frac{\partial}{\partial r}\left(r \cdot \frac{C_1}{r}\right) = \frac{\partial (C_1)}{\partial r} = 0$; $C_1 \neq f(r)$

(c) (50 points)

Use the Navier-Stokes equations to find the velocity profile. Start by simplifying them by using your results from (a) and (b). When solving the resulting differential equation, state and apply appropriate boundary conditions.

In order to be able to find an analytical solution, neglect the nonlinear term $\rho(\underline{v} \cdot \nabla)\underline{v}$, where C is your integration constant from (b)). Write your final expression for the velocity in the box provided below.

Note: Express your final answer in terms of Δp, μ, a, R, h, z, r.

$v_\theta, v_z = 0$ ✓

r : $\rho\left(v_r \frac{\partial v_r}{\partial r}\right) = \frac{-\partial P}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right) + \frac{\partial^2 v_r}{\partial z^2}\right]$

$-C_1(z)/r$; 0 ; Mass balance

θ : $-\frac{1}{r}\frac{\partial P}{\partial \theta} = 0 \Rightarrow \frac{\partial P}{\partial \theta} = 0$ $P \neq f(\theta)$

z : $\frac{-\partial P}{\partial z} + \rho g_z = 0$

$P = f(r, z)$

$P \neq P_x, P_y$; $P = P_r, P_\theta, P_z$

$\frac{\partial P}{\partial z} = \rho g_z$

$P = \rho g z + C_2(r)$

@ BC @ z

From (b) : $v_r = \frac{C_1(z)}{r}$; $\frac{\partial v_r}{\partial r} = -\frac{C_1(z)}{r^2}$; $\frac{\partial (r v_r)}{\partial r} = \frac{\partial}{\partial r}\left(r \cdot \frac{C_1(z)}{r}\right) = 0$

$\rho\left(v_r \cdot \left(\frac{-C_1(z)}{r^2}\right)\right) = -\frac{\partial P}{\partial r} + \frac{\mu}{r}\frac{\partial^2}{\partial z^2}[C_1(z)]$

$\rho\left(\frac{-C_1^2(z)}{r^3}\right) = -\frac{\partial P}{\partial r} + \frac{\mu}{r}\frac{\partial^2}{\partial z^2}(C_1(z))$

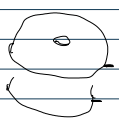
ignore

$\frac{\partial P}{\partial r} = \frac{\mu}{r}\frac{\partial^2 C_1(z)}{\partial z^2}$

Integrate both sides w.r.t r

$\int_{aR}^{R} \frac{\partial P}{\partial r}\, dr = \int_{aR}^{R} \left(\frac{\mu}{r}\frac{\partial^2 C_1(z)}{\partial z^2}\right) dr$

$P(r, z)\Big|_{aR}^{R} = \mu \frac{\partial^2 C_1(z)}{\partial z^2} \int_{aR}^{R} \frac{1}{r}\, dr$



$$P(r=R,z) - P(r=aR,z) = \Delta P = \mu \frac{\partial^2 C_1(z)}{\partial z^2} \ln\left(\frac{R}{aR}\right)$$

$$\frac{\partial^2 C_1(z)}{\partial z^2} = \frac{\Delta P}{\mu} \underbrace{\left[\ln\left(\frac{1}{a}\right)\right]^{-1}}_{-\ln a}$$ Int

$$\frac{\partial C_1(z)}{\partial z} = \left(\frac{\Delta P}{-\mu \ln a}\right) z + A_1$$

$$C_1(z) = \left(\frac{\Delta P}{-\mu \ln a}\right) z^2 + A_1 z + A_2$$

$v_r = \frac{C_1}{r}$ ⇒ @ BC: @ $z=h$ $v_r = 0$ ⇒ $v_r = \frac{1}{r}\left(\lambda h^2 + A_1 h + A_2\right) = 0$

@ $z=-h$ $v_r = 0$ ⇒ $v_r = \frac{1}{r}\left(\lambda h^2 + (-A_1 h) + A_2\right) = 0$

2 eq, 2 unknown

$A_2 = -\frac{\lambda h^2}{2}$, $A_1 = 0$

Δp

$$v_r(r,z) = \frac{\lambda h^2}{2r}\left(\left(\frac{z}{h}\right)^2 - 1\right)$$

Reynold's Transport Theorem

$$\frac{D\Psi}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho\psi\, dV\right) + \iint_{CS} \rho\psi\,(\vec{v}\cdot\hat{n})\, dA$$

Material Derivative | Accumulation | Efflux (in - out)

intensive: $\psi = \frac{d\Psi}{dm}$; extensive: Ψ ; ψ is independent of m

$\Psi = \{m, \vec{P}, E\}$

Mass Balance (Continuity Equation)

$\Psi = m$; $\psi = \frac{d\Psi}{dm} = \frac{dm}{dm} = 1$

$$\frac{Dm}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho\, dV\right) + \iint_{CS} \rho\,(\vec{v}\cdot\hat{n})\, dA$$

$\frac{Dm}{Dt} = 0$

Rate of accumulation of mass w/n CV

Net outward flow of mass across the CS "efflux"

@ SS = 0

- > 0 (Net mass going out)
- < 0 (Net mass going in)
- $= 0$ (Mass w/n CV is constant)

Evaluate at all control surfaces of chosen control volume

For incompressible flow: ρ = constant

∴ For SS, incompressible flow through a CV: $\iint_{CS} \rho\,(\vec{v}\cdot\hat{n})\, dA = \frac{dm}{dt}^{0}$



@ SS, incompressible $\iint_{CS} \rho\,(\vec{v}\cdot\hat{n})\, dA = 0$

$$\rho\iint_{A_1} (\vec{v}\cdot\hat{n})\, dA + \rho\iint_{A_2} (\vec{v}\cdot\hat{n})\, dA = 0$$

$$\rho(-v_1)\iint_{A_1} dA + \rho v_2 \iint_{A_2} dA = 0$$

$$-\rho v_1 A_1 + \rho v_2 A_2 = 0$$

∴ $\boxed{\rho v_1 A_1 = \rho v_2 A_2}$

Irrespective of the flow inside the control volume! (Would work anytime for constant velocity!)

Momentum Balance

$\Psi = m\vec{v}$; $\psi = \frac{d\Psi}{dm} = \frac{d(m\vec{v})}{dm}$

$\psi = \vec{v}\frac{dm}{dm}^{1} + m\frac{d\vec{v}}{dm}^{0}$

$\psi = \vec{v}$; $P = m\vec{v}$

RTT becomes:

$$\frac{D(m\vec{v})}{Dt} = \frac{d}{dt}\left(\iiint_{CV} \rho\vec{v}\, dV\right) + \iint_{CS} \rho\vec{v}\,(\vec{v}\cdot\hat{n})\, dA$$

From Newton's second Law:

$$\Sigma\vec{F} = \frac{D}{Dt}(m\vec{v})$$

$$\therefore \underbrace{\Sigma\vec{F}}_{\text{Sum of forces acting on CV i.e. on the fluid}} = \underbrace{\frac{d}{dt}\left(\iiint_{CV} \rho\vec{v}\, dV\right)}_{\text{Rate of acc. of momentum w/n CV}} + \underbrace{\iint_{CS} \rho\vec{v}\,(\vec{v}\cdot\hat{n})\, dA}_{\text{net rate of momentum efflux from CV}}$$

Convective: 0

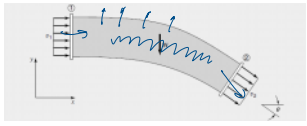
@ SS, convective cont.

$$\Sigma F_x = \iint \rho v_x (\vec{v}\cdot\hat{n})\, dA$$

$$\Sigma F_y = \iint \rho v_y (\vec{v}\cdot\hat{n})\, dA$$

$$\Sigma F_z = \iint \rho v_z (\vec{v}\cdot\hat{n})\, dA$$

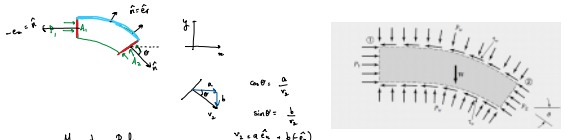
Find the force exerted on a pipe bend resulting from steady state, incompressible flow as shown. W represents gravitational force



④ Define CV → All fluid in the pipe

These are external forces acting on this fluid: pressure, τ_w } Total resultant force is defined $\vec{F}$

Force balance:



$\cos\theta = \frac{a}{v_2}$; $\sin\theta = \frac{b}{v_2}$

$\vec{v}_2 = a\hat{e}_x + b(-\hat{e}_y) = v_2\cos\theta\,\hat{e}_x - v_2\sin\theta\,\hat{e}_y$

Surface	Momentum Balance
① ✓	$\Sigma\vec{F}_1 = \iint_{A_1} \rho\vec{v}(\vec{v}\cdot\hat{n})\, dA_1 = \rho\vec{v}\iint_{A_1} (\vec{v}\cdot\hat{n})\, dA_1 = \rho v_1\hat{e}_x \iint_{A_1} (v_1\hat{e}_x \cdot(-\hat{e}_x))\cdot dA_1 = (-\rho v_1^2 A_1)\hat{e}_x$
③	$\Sigma\vec{F}_3 = \iint_{A_3} \rho\vec{v}(\vec{v}\cdot\hat{n})\cdot dA_3$ → (0)
② ✓	$\Sigma\vec{F}_2 = \iint_{A_2} \rho\vec{v}(\vec{v}\cdot\hat{n})\cdot dA_2 = \rho\vec{v}\iint (\vec{v}_2\cdot\hat{n})\, dA_2 = \rho\vec{v}_2(v_2)\iint dA_2 = \rho v_2 A_2 \begin{pmatrix} v_2\cos\theta \\ -v_2\sin\theta \end{pmatrix} = (\rho v_2^2 A_2\cos\theta)\hat{e}_x - (\rho v_2^2 A_2\sin\theta)\hat{e}_y$

∴ $\Sigma\vec{F} = \Sigma\vec{F}_1 + \Sigma\vec{F}_2$ → Total force acting on the fluid in CV

$$\Sigma\vec{F} = \begin{pmatrix} -\rho v_1^2 A_1 + \rho v_2^2 A_2\cos\theta \\ -\rho v_2^2 A_2\sin\theta \end{pmatrix}_{xy}$$

Since the entire system is stationary → net forces are zero

$\vec{F}_{ext} - \Sigma\vec{F} = 0$

Taking control volume as entire object

$F_{ext}^{x} = P_1 A_1 - P_2 A_2\cos\theta + B_x$ (resultant force on the fluid; don't know the sign)

$F_{ext}^{y} = -W + P_2 A_2\sin\theta + B_y$

$$\vec{F}_{ext} = \begin{pmatrix} P_1 A_1 - P_2 A_2\cos\theta + B_x \\ -W + P_2 A_2\sin\theta + B_y \end{pmatrix}_{xy}$$

$$\begin{pmatrix} P_1 A_1 - P_2 A_2\cos\theta + B_x - (-\rho v_1^2 A_1 + \rho v_2^2 A_2\cos\theta) \\ -W + P_2 A_2\sin\theta + B_y - (-\rho v_2^2 A_2\sin\theta) \end{pmatrix}_{xy} = 0$$

↓

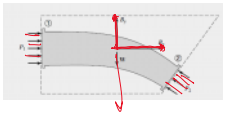
Solve for B_x, B_y

Now force exerted on pipe by fluid will equal and opposite to force exerted by fluid on pipe

$$\underline{F}_{cat} = \begin{pmatrix} P_1 A_1 - \gamma_2 A_2 \cos\theta + B_x \\ -W + P_2 A_2 \sin\theta + B_y \end{pmatrix}_{xy}$$

Now force exerted on pipe by fluid will exert and opposite to force exerted by fluid on pipe

$$R_x = -B_x$$
$$R_y = -B_y$$



Algorithm:

1) Sketch the problem (if no figure provided) ✓
   + Choose coordinate system (or check consistency w/ one provided)
   + Clearly understand where flow is; where the boundaries are

2) Characterize the flow: $\underline{v} = (\,)$; $\underline{g} = (\,)$ ✓
   + Newtonian? Incompressible? Steady state? Where is bulk fluid flowing?
   + Is it open to environment? → Most likely no pressure gradients! ✓
   + Is it inclined? How is gravity affected?

3) Continuity Equation
   + Simplify (and write!) the modified equation (after cancellations)
   + Clearly write why are cancelling out terms

4) Equation of Motion (Microscopic Momentum balance):
   + Incompressible, newtonian (constant viscosity) → Navier stokes ✓
   + Non-newtonian → Cauchy momentum balance

5) Boundary conditions (BC's):
   + No. of BC's correspond to order of differential equation
   + @ solid surfaces → No slip
     @ interfaces → shear stress

6) Solve for velocity profile (and/or pressure field) ✓
   + Integration → keep check of constants
   + solve for constants at very end
   + If pressure is imp (ie. flow in a closed pipe) → then need additional BC's for this

7) Evaluate Q, F given the solved velocity profile



Exam taking strategies

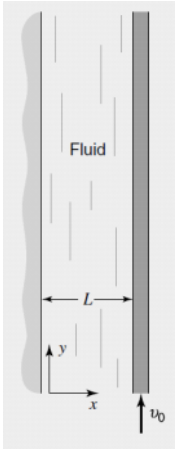
1) Don't panic! → - Read question carefully
   - understand where, what how
   - Check coordinate system !!

2) Maximize points: - Focus on questions w/ higher points (more points ~ more time)
   - Attempt everything & write best guess if short for time (partial credits)

Practice

1) Falling fluid between two plates:

An incompressible fluid confined between two parallel, vertical surfaces. One surface is stationary, whereas the other is moving upward at a constant velocity v0. If we consider the fluid Newtonian and the flow laminar: sketch the velocity streamlines and find the velocity profile in this flow.



+ Cartesian coordinate system (xyz)
+ gravity ✓
+ Newtonian, incomp. ✓
+ Pressure? → closed system so most likely present

$\underline{v} = \begin{pmatrix} 0 \\ -v_y \\ 0 \end{pmatrix}_{xyz}$ $\quad v_y = f(x)$ (Wide/infinite wall i.e. symmetric in z direction)

$\underline{g} = \begin{pmatrix} 0 \\ -g \\ 0 \end{pmatrix}_{xyz}$

→ Mass balance: $0 = \nabla \cdot \underline{v}$ (SS, incomp.)

$\frac{\partial v_x}{\partial x}^{0} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}^{0} = 0 \quad \therefore \frac{\partial v_y}{\partial y} = 0$ [Developed flow]

→ EOM:

$$\rho\left(\overset{SS}{\cancel{\frac{\partial \underline{v}}{\partial t}}} + \underline{v}\cdot\nabla\underline{v}\right) = -\nabla p + \mu\nabla^2\underline{v} + \rho\underline{g}$$

x: $\quad 0 = -\frac{\partial P}{\partial x}$

y: $\quad 0 = -\frac{\partial P}{\partial y} + \mu\left(\frac{\partial^2 v_y}{\partial x^2}\right) + \rho g_y \quad ; \quad g_y = -g$

z: $\quad 0 = -\frac{\partial P}{\partial z}$

No pressure gradients in x & z direction; $\frac{\partial P}{\partial x} = \frac{\partial P}{\partial z} = 0 \qquad P = f(y) \; ; \; \frac{\partial P}{\partial y} = \frac{dP}{dy} = \text{constant}$

$$\therefore \quad \frac{1}{\mu}\underbrace{\left(\frac{dP}{dy} - \rho g\right)}_{\text{Not a function of } x!\ = \lambda} = \frac{\partial^2 v_y}{\partial x^2}$$

$$\frac{\partial v_y}{\partial x} = \lambda x + C_1$$

Not a function of x!
$= \lambda$

$$\frac{\partial v_y}{\partial x} = \frac{\lambda}{\mu} x + C_1$$

$$v_y = \frac{\lambda}{\mu}\frac{x^2}{2} + C_1 x + C_2$$

Boundary conditions:

@ $x = 0$, $v_y = 0$

@ $x = L$, $v_y = v_0$

$C_2 = 0$

$$\left(v_0 - \frac{\lambda}{\mu}\frac{L^2}{2}\right)\frac{1}{L} = C_1$$

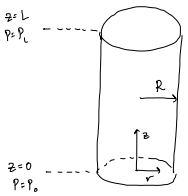
$$C_1 = \frac{v_0}{L} - \frac{\lambda L}{2\mu}$$

$$\boxed{v_y = \frac{\lambda}{\mu}\frac{x^2}{2} + \left(\frac{v_0}{L} - \frac{\lambda L}{2\mu}\right)x}$$

Sketch it out !

2) 6.44 & 6.52

Upward pressure-driven flow in a tube ; Non-newtonian fluid (w/ power law) ; Incompressible ; SS



$\underline{v} = \begin{pmatrix} 0 \\ 0 \\ v_z \end{pmatrix}_{rθz}$

$\underline{g} = \begin{pmatrix} 0 \\ 0 \\ -g \end{pmatrix}_{rθz}$

Flow profile?

Pressure profile?

Cylindrical coordinates

→ Mass balance : $\nabla \cdot \underline{v} = 0$ (incomp., ss.)

$$\cancel{\frac{\partial \rho}{\partial t}}^{0} + \frac{1}{r}\frac{\partial}{\partial r}(\rho r \cancel{v_r}^{0}) + \frac{1}{r}\frac{\partial}{\partial \theta}\cancel{(\rho v_\theta)}^{0} + \frac{\partial}{\partial z}(\rho v_z) = 0$$

$$\frac{\partial v_z}{\partial z} = 0$$

→ Non-newtonian flow → Cannot use Navier stokes!
(only incompressible flows)

Table B.6 : Cylindrical coordinates of Cauchy momentum balance

$\tau = \eta(\dot\gamma)\,\dot{\underline{\gamma}}$

η for non-newt. ↓ ; $\dot{\underline{\gamma}} = \left(\nabla \underline{v} + (\nabla \underline{v})^T\right) = \begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix}$

$\eta(\dot\gamma) = m\dot\gamma^{n-1}$

Equation 8.8-2 :

Newtonian constitutive equation :

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix} = \eta(\dot\gamma) \underbrace{\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{1}{r}\frac{\partial v_z}{\partial \theta} + \frac{\partial v_\theta}{\partial z} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{r\theta z}}_{\dot{\underline{\gamma}}}$$

$$\tau_{rz} = \frac{\partial v_z}{\partial r}$$

$$\dot\gamma = \|\dot{\underline{\gamma}}\| = \left\| \begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial r} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial r} & 0 & 0 \end{pmatrix} \right\|$$

$$= \sqrt{\frac{1}{2}\left(\text{sum of squares of each term in } \dot{\underline{\gamma}}\right)}$$

$$= \sqrt{\frac{1}{2}\left[\left(\frac{\partial v_z}{\partial r}\right)^2 + \left(\frac{\partial v_z}{\partial r}\right)^2\right]} = \left|\frac{\partial v_z}{\partial r}\right|$$

PROBLEM: 6.44 Flow Problem: Upward, pressure-driven flow of a Newtonian fluid in a pipe. Calculate the velocity profile and flow rate for pressure-driven flow of an incompressible, Newtonian liquid in a vertical pipe of radius R. The pressure at the bottom entrance to the tube is P_0; at a point a distance L upward the pressure is P_L. Assume that the flow is well developed and at steady state. Do not neglect gravity.

PROBLEM: 6.52 Flow Problem: Upward pressure-driven flow in a tube, power-law non-Newtonian fluid. Repeat problem 6.44 with a power-law, generalized Newtonian fluid with parameters m and n.

as $r\uparrow$, $v_z\downarrow$ (closer to the wall)

$$\frac{\partial v_z}{\partial r} < 0 \quad \rightarrow \quad -\frac{\partial v_z}{\partial r} > 0$$

$$\therefore \quad \dot{\gamma} = -\frac{\partial v_z}{\partial r}$$

$$\begin{pmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ & \tau_{\theta\theta} & \tau_{\theta z} \\ & & \tau_{zz} \end{pmatrix} = \eta(\dot{\gamma}) \underbrace{\begin{pmatrix} 2\frac{\partial v_r}{\partial r} & r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right) + \frac{1}{r}\frac{\partial v_r}{\partial \theta} & \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r} \\ & 2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}\right) & \frac{\partial v_\theta}{\partial z} + \frac{1}{r}\frac{\partial v_z}{\partial \theta} \\ & & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{rza}}_{\dot{\gamma}}$$

$$= m\dot{\gamma}^{n-1} \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$= m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1} \begin{pmatrix} 0 & 0 & \partial v_z/\partial r \\ & 0 & 0 \\ & & 0 \end{pmatrix}$$

$$\therefore \quad \tau_{rz} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) \neq f(\theta, z)$$

Only τ_{rz} exists (due to flow direction) which is a function of r (and not z!)

r-direction: $0 = -\frac{\partial P}{\partial r}$

θ-direction: $0 = -\frac{1}{r}\frac{\partial P}{\partial \theta}$

z-direction: $0 = -\frac{\partial P}{\partial z} + \left(\frac{1}{r}\frac{\partial}{\partial r}(r\cdot\tau_{rz})\right) + \underbrace{\rho g_z}_{-fg}$

P has no θ & r dependence

$-\frac{\partial P}{\partial z} = -\frac{dP}{dz}$

$$\therefore \quad \underbrace{\frac{dP}{dz} + \rho g}_{\neq f(r)} = \underbrace{\frac{1}{r}\frac{d}{dr}(r\tau_{rz})}_{\neq f(z)} = \lambda$$

↳ Solve them separately!

↳ $\frac{dP}{dz} + \rho g = \lambda$

$\frac{dP}{dz} = \lambda - \rho g$

$P = (\lambda - \rho g)z + c_1$

@ $z=0$, $P = P_0$; @ $z=L$, $P = P_L$ } $c_1 = P_0$

$(P_L - P_0) = (\lambda - \rho g)L$

$\lambda = \frac{P_L - P_0}{L} + \rho g$

↳ $\frac{d}{dr}(r\tau_{rz}) = \lambda r$

$\tau_{rz} = \frac{\lambda r}{2} + \frac{c_1}{r}$

@ $r=0$; $\frac{\partial v_z}{\partial r} = 0$; $\tau_{rz} = 0$ } $c_1 = 0$

$\tau_{rz} = \frac{\lambda r}{2}$

↳ $\tau_{rz} = m\left(-\frac{\partial v_z}{\partial r}\right)^{n-1}\left(\frac{\partial v_z}{\partial r}\right) \times \frac{(-1)}{(-1)}$

$\tau_{rz} = -m\left(-\frac{\partial v_z}{\partial r}\right)^n$

$\tau_{rz} = \frac{\lambda r}{2} = -m\left(-\frac{\partial v_z}{\partial r}\right)^n$

$\downarrow$

$-\frac{dv_z}{dr} = \left(-\frac{\lambda}{2m}\right)^{\frac{1}{n}} r^{1/n}$

$v_z = -\left(-\frac{\lambda}{2m}\right)^{1/n} \frac{r^{1/n+1}}{\left(\frac{1}{n}+1\right)} + c_1$

@ $r=R$, $v_z = 0$ } $c_1 = \left(-\frac{\lambda}{2m}\right)^{1/n} \frac{R^{(\frac{1}{n}+1)}}{\left(\frac{1}{n}+1\right)}$

$\therefore$ $\boxed{v_z = \left(-\frac{\lambda}{2m}\right)^{1/n} \frac{1}{\left(\frac{1}{n}+1\right)}\left(R^{\frac{1}{n}+1} - r^{\frac{1}{n}+1}\right)}$

Review session MT3

Tuesday, 28 November 2023 9:14 AM

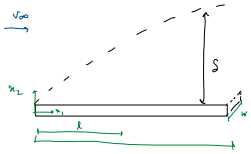
Boundary Layer

Laminar → Transitional → Turbulent

$Re_L = > 1 \times 10^6$

- Laminar $[Re_{x_1} < 2 \times 10^5]$
  - + No slip condition (gives rise to flow dependent on two directions)
- Turbulent $[Re_{x_1} > 1 \times 10^6]$
  - + Downstream flow (from the leading edge) the velocity becomes unstable } i.e. only starts after a initial laminar regime BL
  - + Thicker layers (that have different x_1 dependence) } Higher drag on flat plate

$C_D = \frac{F_{drag}}{(KE)(Area)}$ ✓



Smooth flat plate BL

$\delta = \left(\frac{25\mu}{\rho v_\infty}\right)\sqrt{x} = \frac{5x}{\sqrt{Re}}$

distance from leading edge of the plate [Not a var!]

$\delta = \frac{0.16x}{(Re)_x^{1/7}}$; $C_D = \frac{0.031}{(Re_L)^{1/7}}$; $C_f = \frac{0.027}{(Re_x)^{1/7}} = \frac{2\tau_w}{\rho U_0^2}$; $U_0 = v_\infty$

$\tau_w = \frac{0.0135\,\rho v_\infty^2}{(Re_x)^{1/7}}$

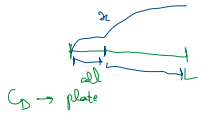
Fluid force on plate:

$F = \iint_S [\hat{n} \cdot \Pi]|_{surf}\, dS$ → pg 693 → $\tilde{\tau}_w = 0.332\,\mu v_\infty \left(\frac{\rho v_\infty}{\mu x}\right)^{1/2}$

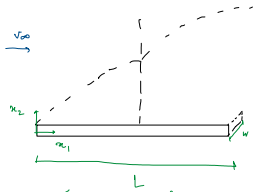
$F_{drag} = \int_0^W \int_0^L \tilde{\tau}_w\, dx_1\, dx_3$; $Re = \frac{\rho v_\infty x}{\mu}$

★

$C_D = \frac{F_{drag}}{\frac{1}{2}\rho v_\infty^2 WL} = \frac{1.328}{(Re)_x^{1/2}}$



$C_D \rightarrow$ plate



Smooth flat plate BL

Laminar (upto a certain x_1) | turbulent

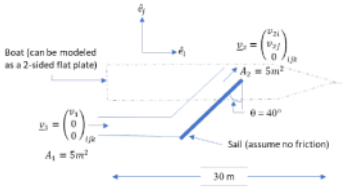
General relationship: $C_D = \frac{F_{drag}}{(KE)(Area)} = \frac{\iint \tau_w \cdot dA}{(KE)(Area)} = \frac{\iint C_F \cdot \left(\frac{\rho v_\infty^2}{2}\right) \cdot dA}{(KE)(Area)}$

dependent on lam/turb } dependent on x var

} Use C_F Unless evaluating for entire plat BL!)

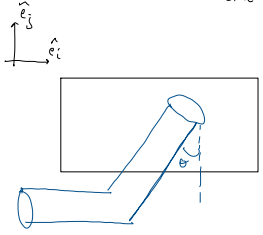
1. Come Sail Away

You are captain on a sailboat. Your boat is moving completely due to the force of the wind in the sail. The wind enters the sail region through an area that is roughly 5 m², with a velocity $\underline{v}_3 = \begin{pmatrix} v_3 \\ 0 \\ 0 \end{pmatrix}_{ijk}$. The wind leaves the sail region through an area that is also 5 m², with a velocity $\underline{v}_2 = \begin{pmatrix} v_{2x} \\ v_{2y} \\ 0 \end{pmatrix}_{ijk}$. Your boat is 30m long, and it can be modeled as a 2-sided flat plate. You can consider that the boat extends 2 m below the surface of the water. On this day, your boat is moving at a constant velocity of 0.5 m/s. The water has a density of 1000 kg/m³ and a viscosity of 9x10⁻³ poise. The viscosity of air is 1.7x10⁻⁵ poise, and the density of air is 1.2 kg/m³.

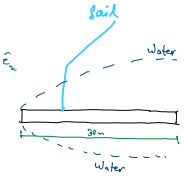


[Boat]

Two-sided flat plate;

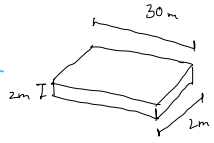


Side view of the boat: →



"Kinda like a hydrofoil of a sailboat" I.e. submerged.





1 poise = 0.1 $\frac{kg}{m \cdot s}$

Water BL: $Re_L = \frac{\rho v_\infty L}{\mu}$

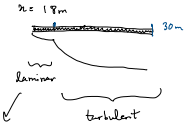
$= \frac{(1000)(0.5)(30)}{1.7 \times 10^{-7}}$ (turbulent)

Water BL: $Re_L = \frac{\rho v_\infty L}{\mu}$

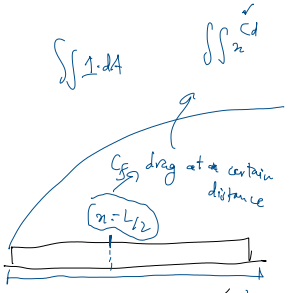
$= \frac{(1000)(0.5)(30)}{(0.9)\times 10^{-3}} = 1.67\times 10^7$ (turbulent)

Finding transition to turbulent: $1\times 10^6 = \frac{(1000)(0.5)(L)}{(0.9\times 10^{-3})}$

$L = 1.8$ m



$\tau_w^l = \frac{0.332\,\rho v_\infty^2}{(Re_x)^{1/2}}$ $\quad \tau_w^t = \frac{0.0135\,\rho v_\infty^2}{(Re_x)^{1/7}}$



$C_D = f(Re_L)$ ✓ on entire plate

$F_D = \iint \tau_w^l \cdot dx\,dy + \iint \tau_w^t \cdot dx\,dy$

$= \int_0^2\int_0^{1.8} \frac{(0.332)\,\rho v_\infty^2}{\left(\frac{\rho v_\infty x}{\mu}\right)^{1/2}} dx\,dy + \int_0^2\int_{1.8}^{30} \frac{0.0135\,\rho v_\infty^2}{\left(\frac{\rho v_\infty x}{\mu}\right)^{1/7}} \cdot dx\,dy$

$\frac{2-\frac{1}{7}}{\frac{1}{2}-\frac{1}{7}} = \frac{13}{5}$ $\quad -\frac{1}{7}+\frac{1}{1+7}$

$= 2\left(\int_0^{1.8} 0.332\,\rho^{1/2} v_\infty^{3/2} \mu^{1/2} x^{-1/2} \cdot dx\right) + 2\left(\int_{1.8}^{30} 0.0135\,\rho^{6/7} v_\infty^{13/7} \mu^{1/7} x^{-1/7}\, dx\right)$

$= 2(0.332)\sqrt{\rho\mu}\,(v_\infty)^{3/2} \frac{x^{1/2}}{(1/2)}\Big|_0^{1.8} + 2(0.0135)\,\rho^{6/7} v_\infty^{13/7} \mu^{1/7} \frac{x^{6/7}}{(6/7)}\Big|_{1.8}^{30}$

$F_D = \left[2(0.332)\sqrt{(1000)(0.9\times 10^{-3})}\cdot(0.5)^{3/2}\left(\frac{(1.8)^{1/2}}{(1/2)}\right)\right] + \left[2(0.0135)(1000)^{6/7}(0.5)^{13/7}(0.9\times 10^{-3})^{1/7}\,\frac{30^{6/7}-(1.8)^{6/7}}{(6/7)}\right]$

$= \underbrace{0.5976}_{0.6\text{ N}} + \underbrace{19.99}_{20\text{ N}}$

Force of drag from just 1 BL

$F_D = 20.6$ N

Total F_D (from both BL) $\Rightarrow 2F_D = F_D^{total}$ $\Rightarrow$ $\boxed{F_D^{total} = 41.2\text{ N}}$

Evaluate the velocities using momentum balance:

$\beta = 1$ (turbulent) ; $\frac{d\vec{p}}{dt}^{\to 0} + \sum \rho \cos\theta_i A_i \langle v_i\rangle^2 \hat{v}_i = \sum \vec{f} = \underbrace{\vec{f}_g + \vec{f}_{in} + \vec{f}_{out}}_{\text{negligible}} + \vec{f}_w$; $f_w = -F_D^{total}$

$\rho\cos\theta_1 A_1 \langle v_1\rangle^2 \hat{v}_1 + \rho\cos\theta_2 A_2 \langle v_2\rangle^2 \hat{v}_2 = -41.2$ N

$\rho A\left[(-1)\langle v_1\rangle^2 \hat{e}_n + (1)\langle v_2\rangle^2 \begin{pmatrix}\cos 50\\ \sin 50\end{pmatrix}\right] = -41.2$ N

{ only care about i^{th} direction:

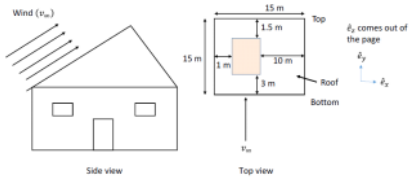
$\Rightarrow \rho A\left[-\langle v_1\rangle^2 + \langle v_2\rangle^2 \cos 50\right] = -41.2$; $v_1 = v_2 = (v_1 - 0.5)$ ← Finding the relative velocity (since boat is moving as well)

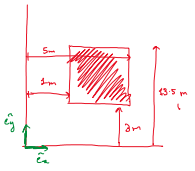
$\Rightarrow \rho A\left[-(v_1-0.5)^2 + (v_1-0.5)^2\cos 50\right] = -41.2$

$(v_1-0.5)^2 \rho A\,(-1+\cos(50)) = -41.2$

$\boxed{v_1 = 4.88\ m/s}$

Raise the Roof





Given an Fdrag $F_D = C_f \cdot F_g$; $F_g = m \cdot g$

$16 = 3+2+1.5$
$x = 5+4.5$
$= 10.5$

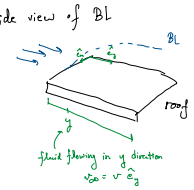
We don't know the v_∞ → have to assume a regime since we cannot find Re

Assuming turbulent on the board (i.e. laminar regime has been crossed before the board ↓ But we don't care)

$$F_D = \iint \tau_w \cdot dx\, dy \quad ; \quad \tau_w = \frac{\rho v_\infty^2}{2} C_f = \frac{0.0135\, \rho v_\infty^2}{(Re_x)^{1/7}} \quad ; \quad Re_y = \frac{\rho v_\infty y}{\mu}$$

Boundary layer is developing in the y direction → that is the characteristic dimension now (given the coordinate system we have)

Side view of BL



↳ Given exam solution are also technically correct but they changed the coordinate axis given arbitrarily

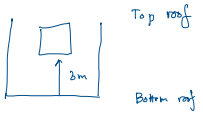
$$F_D = \int_3^{13.5} \int_1^5 0.0135\, \rho^{6/7} v_\infty^{13/7} \mu^{1/7} (y)^{-1/7}\, dx\, dy$$

$$= (5) \left[0.0135\, \rho^{6/7} v_\infty^{13/7} \mu^{1/7} \frac{(y)^{(6/7)}}{(6/7)} \right]_3^{13.5}$$

$$C_f \cdot (F_g) = v_\infty^{13/7} \left[(5)(0.0135) \left[\frac{1}{3} \right]^{6/7} (1.7 \times 10^{-5})^{1/7} \left[\frac{13.5^{6/7} - 3^{6/7}}{6/7} \right] \right]$$

$$(1.2)(70) = v_\infty^{13/7} (0.1385)$$

$$v_\infty = \left(\frac{(1.2)(70)}{0.1385} \right)^{7/13}$$



$$\boxed{v_\infty = 31.5 \approx 32 \text{ m/s}}$$

↳ Assumption check: @ start of board ↓

$$Re = \frac{(1.3)(31.5)(3)}{(1.7 \times 10^{-5})} > 1 \times 10^6$$

✓ Board is in turbulent

$\delta^{turb} = \frac{0.16 \cdot x}{(Re_x)^{1/7}}$ → x @ which we want to measure the thickness (δ changes w/ x!)

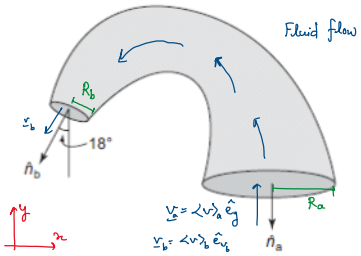
∴ @ x = 14.5 (the birds nest)

$$\delta^{turb} = \frac{(0.16)(14.5)}{\left(\frac{\rho v_\infty (14.5)}{\mu} \right)^{1/7}} = \frac{(0.16)(14.5)}{\left[\frac{(1.3)(31.5)(14.5)}{(1.7 \times 10^{-5})} \right]^{1/7}} = 0.194 \text{ m}$$

Nest is 0.25 m tall ; $\delta_{turb} = 0.19$ m

It is not in the BL & experiences v_∞ (will likely 'blow away)!

Question 1



Turbulence
↓
Time-averaged quantity
✓ Steady state ✓

$$\overline{\text{quantity}} = \int_{t_0}^{t_1} \text{quantity} \cdot dt \qquad \frac{d\bar{v}}{dx}$$

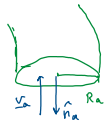
A liquid with a given viscosity (mew) enters and leaves the above bend as shown. Solve the following questions

(a) Using Reynolds transport theorem, solve for the overall force on the fluid.

$$\text{RTT}: \quad \frac{d\underline{P}}{dt} + \iint_{CS} \rho (\underline{v} \cdot \hat{n}) \underline{v} \, dS = \sum_{on\ CV} \underline{F}$$

Taking the fluid inside the pipe as the CV : there is flow through surface A and surface B

Inlet:



$$\underline{v}_a = \begin{pmatrix} 0 \\ \langle v \rangle \\ 0 \end{pmatrix}_{xyz} \quad ; \quad \hat{n}_a = -\hat{e}_y = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}_{xyz}$$

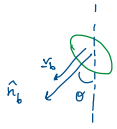
$$\iint_{inlet} \rho \underline{v}_a (\hat{n} \cdot \underline{v}_a) \, dS = \int_0^{2\pi} \int_0^{R_a} \rho (\langle v \rangle_a \hat{e}_y) \left[\begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}_{xyz} \cdot \begin{pmatrix} 0 \\ \langle v \rangle_a \\ 0 \end{pmatrix}_{xyz} \right] r \, dr \, d\theta$$

$$= (\rho \langle v \rangle_a \hat{e}_y)(-\langle v \rangle_a) \int_0^{2\pi} \int_0^{R_a} r \, dr \, d\theta$$

$$= (-\rho \langle v \rangle_a^2 \hat{e}_y) \left(2\pi \cdot \frac{r^2}{2} \Big|_0^{R_a} \right)$$

$$= (-\rho \langle v \rangle_a^2 \pi R_a^2) \hat{e}_y = \begin{pmatrix} 0 \\ -\rho \langle v \rangle_a^2 \pi R_a^2 \\ 0 \end{pmatrix}_{xyz}$$

Outlet:





$$\hat{n}_b = \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz} \qquad \hat{e}_{v_b} = \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz} \quad ; \quad \underline{v}_b = \langle v \rangle_b \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz}$$

$$\iint_{outlet} \rho \underline{v}_b (\hat{n} \cdot \underline{v}_b) \cdot dS = \int_0^{2\pi} \int_0^{R_b} \rho (\langle v \rangle_b \hat{e}_{v_b}) \left[\begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz} \cdot \begin{pmatrix} -\langle v \rangle_b \sin\theta \\ -\langle v \rangle_b \cos\theta \\ 0 \end{pmatrix}_{xyz} \right] r \, dr \, d\theta$$

$$= \rho [\langle v \rangle_b \hat{e}_{v_b}] \left[\langle v \rangle_b \sin^2\theta + \langle v \rangle_b \cos^2\theta + 0 \right] \int_0^{2\pi} \int_0^{R_b} r \, dr \, d\theta$$

$$= \rho (\langle v \rangle_b \hat{e}_{v_b}) (\langle v \rangle_b) (\pi R_b^2)$$

$$= \rho \langle v \rangle_b^2 \pi R_b^2 \hat{e}_{v_b}$$

$$= \left[\rho \langle v \rangle_b^2 \pi R_b^2 \right] \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz}$$

Overall RTT @ steady state:

$$\underset{Con\ W}{\sum \underline{F}} = \iint_{CS} \rho (\underline{v} \cdot \hat{n}) \underline{v} \, dS = \begin{pmatrix} 0 \\ -\rho \langle v \rangle_a^2 \pi R_a^2 \\ 0 \end{pmatrix}_{xyz} + \left[\rho \langle v \rangle_b^2 \pi R_b^2 \right] \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}_{xyz} = \boxed{\begin{pmatrix} -\rho \langle v \rangle_b^2 \pi R_b^2 \sin\theta \\ -\rho \langle v \rangle_a^2 \pi R_a^2 - \rho \langle v \rangle_b^2 \pi R_b^2 \cos\theta \\ 0 \end{pmatrix}_{xyz}}$$

(b) Using macroscopic momentum balance, solve for the overall force on the fluid

@SS

Macro Momentum balance: $\sum_{\text{on CV}} \underline{F} = \frac{d\underline{P}}{dt} + \sum_{i=1}^{\#} \left[\frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right]$

Inlet: $\theta_a = 180^\circ$ Outlet: $\theta_b = 0^\circ$

$$\sum_{\text{(on CV)}} \underline{F} = \left. \frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right|_{\text{inlet}} + \left. \frac{\rho A \cos\theta \langle v \rangle^2}{\beta} \hat{v} \right|_{\text{outlet}}$$

$$\Rightarrow \sum_{\text{on CV}} \underline{F} = \frac{\rho A_a \cos\theta_a \langle v \rangle_a^2}{\beta} (\hat{e}_y) + \frac{\rho A_b \cos\theta_b \langle v \rangle_b^2}{\beta} \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$$

$$\Rightarrow \sum_{\text{on CV}} \underline{F} = \frac{\rho \pi R_a^2 \cdot (-1) \langle v \rangle_a^2}{\beta} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{\rho \pi R_b^2 (1) \langle v \rangle_b^2}{\beta} \begin{pmatrix} -\sin\theta \\ -\cos\theta \\ 0 \end{pmatrix}$$

For turbulent flow: $\beta = 1$

$$\sum_{\text{on CV}} \underline{F} = \begin{bmatrix} -\rho \pi R_b^2 \langle v \rangle_b^2 \sin\theta \\ -\rho \pi R_a^2 \langle v \rangle_a^2 - \rho \pi R_b^2 \langle v \rangle_b^2 \cos\theta \\ 0 \end{bmatrix}$$

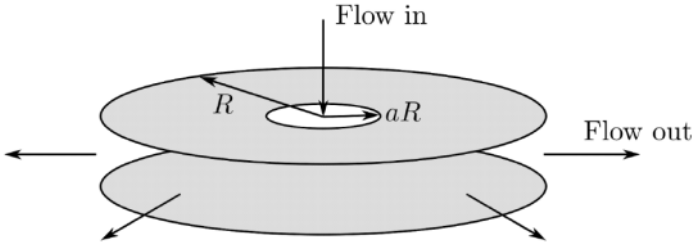
(c) Are your force expressions in (a) and (b) different from each other? Why or why not?

No! Because RTT & MMB are the one & same → macroscopic look @ CV

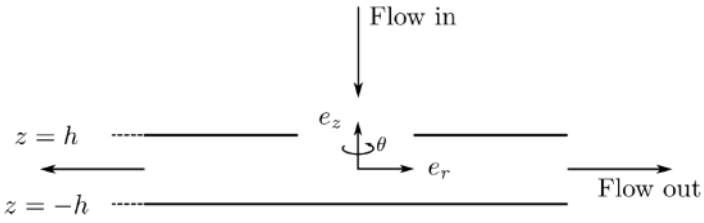
↳ Using either to arrive at expression are valid !

Question 2

Consider two concentric disks which are fixed to stay in place. There is an inflow of an incompressible fluid through a hole in the top plate. The flow is driven by a pressure drop, $\Delta p = p(r = aR) - p(r = R)$, between the inlet at $r = aR$ ($a \ll 1$) and the outlet at $r = R$. Below, you will find the 3D view of the apparatus.



The following figure shows the front view of the apparatus above. Consider the cylindrical coordinate system to be positioned in the center.



You can make the following assumptions:

1. The outflow is homogeneous along the circumference.
2. The flow is fully developed and at steady-state.
3. The fluid is incompressible.
4. The outlet is at atmospheric pressure p_{atm}.
5. The effects due to flow around the corner at the inlet can be neglected ($a \ll 1$). The pressure at that point is unperturbed by the flow around the corner.

(a) (10 points)
Write down the kinematics for this problem by stating physically reasonable assumptions.
(i) Which velocity components are non-zero?
(ii) Which coordinates (i.e. r, θ, z) do the non-zero velocity component(s) depend on?

Kinematics refer to velocity profiles

(i) $\underline{v} = \begin{pmatrix} v_r \\ v_\theta \\ v_z \end{pmatrix}$; $v_\theta = v_z = 0$; only v_r is non-zero

(ii) $v_r(r, \theta, z)$ — symmetric

(b) (20 points)
Use the local form of the balance of mass to determine the radial component of the velocity profile in terms of an integration constant. Write your final answer in the box provided below.

Mass balance : $\nabla \cdot \underline{v} = 0$ (@ SS, incompressible)

$$\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta)^{0} + \frac{\partial}{\partial z}(\rho v_z)^{0} = 0$$

$$\frac{\partial}{\partial r}(\rho r v_r) = 0$$

$$r v_r = 0 \quad \Big\} \text{ Integrate once}$$

$$\boxed{v_r = \frac{C_1}{r}}$$

C_1 has to be function z ($C_1(z)$)
(since we know $v_r(r, z)$ & C_1 is a constant w.r.t variable r)

(c) (50 points)
Use the Navier-Stokes equations to find the velocity profile. Start by simplifying them by using your results from (a) and (b). When solving the resulting differential equation, state and apply appropriate boundary conditions.
In order to be able to find an analytical solution, neglect the nonlinear term ($\rho C^2/r^3$, where C is your integration constant from (b)). Write your final expression for the velocity in the box provided below.

Note: Express your final answer in terms of Δp, μ, a, R, h, z, r.

r-direction:

$$\rho\left(v_r \frac{\partial v_r}{\partial r}\right) = -\frac{\partial P}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right) + \frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction:

$$-\frac{1}{r}\frac{\partial P}{\partial \theta} = 0$$

z-direction:

$$-\frac{\partial P}{\partial z} + \rho g_z = 0$$

r-direction

$$\rho\left(\frac{\partial v_r}{\partial t} + v_r\frac{\partial v_r}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z\frac{\partial v_r}{\partial z}\right) = -\frac{\partial P}{\partial r} + \rho g_r + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2}\right]$$

θ-direction

$$\rho\left(\frac{\partial v_\theta}{\partial t} + v_r\frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta} + \frac{v_r v_\theta}{r} + v_z\frac{\partial v_\theta}{\partial z}\right) = -\frac{1}{r}\frac{\partial P}{\partial \theta} + \rho g_\theta + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_\theta)\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2}\right]$$

z-direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_r\frac{\partial v_z}{\partial r} + \frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$$

$= -\frac{\partial P}{\partial z} + \rho g_z + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right]$

z-direction:

$$-\frac{\partial P}{\partial z} + \rho g_z = 0$$

From part(b): $v_r = \frac{C_1(z)}{r}$; $\frac{\partial v_r}{\partial r} = -\frac{C_1(z)}{r^2}$

Not a function of r!

; $\frac{\partial}{\partial r}(r \cdot v_r) = \frac{\partial}{\partial r}\left(r \cdot \frac{C_1(z)}{r}\right) = 0$

r direction:

$\therefore \quad \rho\left(v_r \frac{\partial v_r}{\partial r}\right) = -\frac{\partial P}{\partial r} + \mu\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial}{\partial r}(r v_r)\right) + \frac{\partial^2 v_r}{\partial z^2}\right]$ (the bracketed term $\to 0$)

$$\rho\left(\frac{C_1(z)}{r} \cdot \left(-\frac{C_1(z)}{r^2}\right)\right) = -\frac{\partial P}{\partial r} + \mu\frac{\partial^2}{\partial z^2}\left(\frac{C_1(z)}{r}\right)$$

$$\Rightarrow \quad \rho\left(-\frac{C_1^2(z)}{r^3}\right) = -\frac{\partial P}{\partial r} + \frac{\mu}{r}\frac{\partial^2}{\partial z^2}(C_1(z))$$

Can neglect as per question

$$\frac{\partial P}{\partial r} = \frac{\mu}{r}\frac{\partial^2}{\partial z^2}(C_1(z))$$

From the other dimensions:

$P = f(r, z)$; $-\frac{\partial P}{\partial z} + \rho g_z = 0$

$\therefore \quad \frac{\partial P}{\partial r} = \frac{\mu}{r}\frac{\partial}{\partial z^2}(C_1(z))$

$\frac{\partial P}{\partial z} = \rho g_z$

$P = \rho g_z z + C_2(r)$

↳ We would need BC's for P in z to solve for $C_2(r)$

Integrating both sides w.r.t r

$$\int_{aR}^{R}\frac{\partial P}{\partial r}\cdot dr = \int_{aR}^{R}\frac{\mu}{r}\frac{\partial}{\partial z^2}(C_1(z))\,dr$$

$$\int_{aR}^{R}\frac{\partial P}{\partial r}\cdot dr = \mu\frac{\partial}{\partial z^2}(C_1(z))\int_{aR}^{R}\frac{1}{r}\cdot dr$$

$$P(r,z)\Big|_{aR}^{R} = \mu\frac{\partial}{\partial z^2}(C_1(z))\,\ln r\Big|_{aR}^{R}$$

$\Delta p = P(r=R) - P(r=aR)$ @ any z

BC's for P:

- @ $r = aR$: $P = P(r=aR) = P_{in}$
- @ $r = R$: $P = P(r=R) = P_{out}$

$$\Delta p = \mu\frac{\partial}{\partial z^2}(C_1(z))\,\ln\left(\frac{R}{aR}\right)$$

$$\frac{\partial}{\partial z^2}C_1(z) = \frac{\Delta p}{\mu}\cdot\left(\ln\left(\frac{R}{aR}\right)\right)^{-1} = \lambda$$

Integrating twice w.r.t z

$$\frac{\partial}{\partial z}C_1(z) = \lambda z + A_1$$

$$C_1(z) = \lambda\frac{z^2}{2} + A_1 z + A_2$$

$$\boxed{v_r = \frac{C_1(z)}{r} = \frac{\lambda z^2}{2r} + \frac{A_1 z}{r} + \frac{A_2}{r}}$$

BC's:

@ $z = h$, $v_r = 0$ $\Rightarrow$ $0 = \frac{1}{r}\left(\frac{\lambda h^2}{2} + A_1 h + A_2\right)$

@ $z = -h$, $v_r = 0$ $\qquad 0 = \frac{1}{r}\left(\frac{\lambda h^2}{2} + A_1(-h) + A_2 \right)$

2 eq ; 2 unknowns : $\qquad 2A_2 = -\lambda h^2$

$$A_2 = -\frac{\lambda h^2}{2} \quad ; \quad A_1 = 0$$

$$\therefore v_r(r,z) = \frac{1}{r}\left(\frac{\lambda z^2}{2} - \frac{\lambda h^2}{2} \right) \quad ; \quad \lambda = \frac{\Delta p}{\mu}\underbrace{\left(\ln\left(\frac{1}{a}\right)\right)^{-1}}_{-\ln a} = \frac{\Delta p}{-\mu(\ln a)}$$

$$v_r(r,z) = \frac{\lambda h^2}{2r}\left(\left(\frac{z}{h}\right)^2 - 1 \right)$$

$$\boxed{v_r(r,z) = \frac{\Delta p h^2}{2\mu r(\ln a)}\left(1 - \left(\frac{z}{h}\right)^2 \right)}$$